

TACTICS AND DUTIES
FOR
TRENCH FIGHTING

GEORGES BERTRAND
AND
OSCAR N. SOLBERT

TACTICS AND DUTIES
FOR
TRENCH FIGHTING

GEORGES BERTRAND
AND
OSCAR N. SOLBERT

The Project Gutenberg eBook of Tactics and duties for trench fighting

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Tactics and duties for trench fighting

Author: Georges Etienne Bertrand
Oscar N. Solbert

Release date: April 20, 2019 [eBook #59317]

Language: English

Other information and formats: www.gutenberg.org/ebooks/59317

Credits: E-text prepared by Richard Tonsing, Brian Coe, and the Online Distributed Proofreading Team (<http://www.pgdp.net>) from page images generously made available by Internet Archive (<https://archive.org>)

***** START OF THE PROJECT GUTENBERG EBOOK TACTICS AND
DUTIES FOR TRENCH FIGHTING *****

**E-text prepared by Richard Tonsing, Brian Coe,
and the Online Distributed Proofreading Team
(<http://www.pgdp.net>)
from page images generously made available by
Internet Archive
(<https://archive.org>)**

Note: Images of the original pages are available through Internet Archive. See
<https://archive.org/details/tacticsdutiesfor00bertrich>

Tactics and Duties for Trench Fighting

By
Georges Bertrand
Capitaine, Chasseurs Alpins, de l'Armée de France

and

Oscar N. Solbert
Major, Corps of Engineers, U.S.A.

With Diagrams

G. P. PUTNAM'S SONS
NEW YORK AND LONDON
The Knickerbocker Press
1918

COPYRIGHT, 1918
BY
OSCAR N. SOLBERT

The Knickerbocker Press, New York

FOREWORD

It must be borne in mind that this book deals with principles and examples of methods of warfare, and not fixed rules. There are no fixed methods in warfare. Development in warfare means change of methods or the invention of new ones to overcome military difficulties. However, one must have knowledge of past and present methods to be ready for the next logical step of development.

HEADQUARTERS, FORT SHERIDAN TRAINING CAMP, ILL.,

October 26, 1917.

From: MAJOR O. N. SOLBERT, C. of E.

To the Adjutant General, U. S. Army, through official channels.

Subject: Publication of Lectures.

I. Request authority to publish in pamphlet form lectures given at this camp on Trench Fighting.



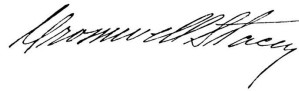
Major, C. of E.

1st Indorsement

OFFICE OF THE SENIOR INSTRUCTOR, FORT SHERIDAN TRAINING CAMP, ILLINOIS,
OCTOBER 26, 1917.

TO THE COMMANDING OFFICER:

Earnestly recommending that the lectures referred to above be published in pamphlet form. I consider it most desirable that every graduate of this camp take these lectures with him, upon being commissioned. I do not believe that the best interests of the service will suffer, in any way by publishing these lectures.



Lt. Colonel, Infantry,
Senior Instructor.

2nd Indorsement

HQ., THE FORT SHERIDAN TRAINING CAMP, OCTOBER 26, 1917.

TO THE ADJUTANT GENERAL OF THE ARMY, WASHINGTON, D. C.

I. Approved.



Colonel of Cavalry,
Commanding.

WAR DEPARTMENT, A. G. O., OCTOBER 31, 1917.

TO HEADQUARTERS, THE FORT SHERIDAN TRAINING CAMP, FORT SHERIDAN, ILL.

Returned.

I. Approved.

By order of the Secretary of War:

A handwritten signature in black ink, appearing to read "T. R. Hays", with a large checkmark at the end.

Adjutant General.

CONTENTS

Chapter I

Organization of the Company for Battle Formations

PAGE

Specialists—Organization of a Typical Company—**Use of Specialists**—In the Defensive—In the Offensive—**Formations**—Initial Formation—Close Order and March Formation—Approach March—Skirmisher Line—Assaulting Formations—Composition of Lines—**Drill Regulations**

[1](#)

Chapter II

Development of a Position from an Open Warfare Battle

General Use of Fortifications—Change from Battle Lines to Trenches of a Position—**Division of a Position**—Different Lines of a Sector—**Principles of Organization**—Flanking Fire—Field of Fire—Location of Trenches—Compartments—Concealment—**Elements of Organization**—Plan—Outline of First Line—Cover Trench—Advance Posts—Loopholes—Traverses—Transversals—Support Trenches and Redoubts—Boyaux and Communication Ditches—Accessory Defenses—Command Posts

[34](#)

Chapter III

Sector and Trench Duties

Plan of Defense—Definition—Principles—Form of—**Details of Trench Duty**—Guard Duties—Sentinels—Reconnoitering Patrols—Observers—Attrition—Fighting Patrols and Raids—Arms, Equipment, and Ammunition—Activities of the Troops—Reports—**Use of Infantry and Artillery Weapons, and Liaison**—Machine Guns—Trench

[62](#)

Mortars—**Artillery Support**—Use—Barrage—Liaison—Instructions for Sentinels—
Watchers—Observers

Chapter IV

The Relief

Definitions and Principles—Kinds—**Preliminary Dispositions for the Relief**—Time to
Prepare—Preparation—Reconnaissance—**Relief Orders**—General—Battalion—
Movement of the Relief—Departure—March to Rendezvous of Guides—Guides—
March in the Boyaux—**Occupation of the Position**—Duties after

[104](#)

Chapter V

Defense of a Position

Hostile Attacks—Kinds—**Surprise**—Precautions against—Maintenance of Barbed Wire
Entanglements—Service of Guard and Observation—“Stand To” Exercises—**Attack in
Force**—Revealing Symptoms—Preventive Measures before—Perfecting Organization
of Position—Increase of Control and Discipline of Defense—Moral Preparation of the
Troops—Maintenance of the Garrison during the Bombardment—Reply with Artillery
Fire—**Defense of First Line during Assault**—Duties of Watchers—Defenders of the
First Line—**Fighting in the Interior of the Position**—Defense of an Area—The
Counterattack—Troops Detailed for—Routes or Directions—Form of—**Repair of
Position**

[123](#)

Chapter VI

Attack of a Position

Phases—Preparation—Assault against First Hostile Line—Exploitation of Success—
Preparation—Front of Attack—**Conditions of Assault**—Distance of—Trace of Lines
of Departure—Ground Preparation—**Organization of Assault**—**Execution of the
Works**—**Artillery Preparation**—Counter-Battery Fire—Fire on Communication
Routes, etc.—Destructive Fire—**Plan of Action**—Observation—Control—**Plan of
Battle**—Strength of Enemy—Preparation of Men for Attack—Mission of Attack—**The
Assault**—Disposition in Depth—Waves—Assaulting Companies—Supporting
Companies—Disposition of Assaulting Battalion in the Departure Trenches—Order of
Attack—Departure of Assault—Support Artillery—Departure of First Echelon—
Departure of Second Echelon—Advance of Reserves—**Fighting in the Interior of a
Position**—Principle of—Details of—Assaulting Companies—Reinforcing Companies

[152](#)

—**Rôle of the Artillery during the Attack**—Principle of Accompaniment Fire—Means of Controlling Barrage—**Liaisons during Attack**—**Aviation**

Chapter VII

Model of Trench Orders

Plan of Defense—Order for Relief—**Plan of Attack**—Order for Attack—Order for a Raid [196](#)

Chapter VIII

Special Operations

Raids—Preparation—Purpose of—Troops Employed—Objective of—Execution of—Infantry Support—Artillery Support—**Gas Warfare**—Kinds—Protections against—Special Precautionary Measures—**Liquid Fire**—**Mines** [216](#)

Tactics and Duties for Trench Fighting

CHAPTER I
ORGANIZATION OF THE COMPANY FOR
BATTLE FORMATIONS

I. Specialists

The exigencies of modern warfare, especially of trench warfare, have developed new infantry weapons. As there are several kinds of these weapons to each infantry company, the men detailed to handle them must be trained as specialists. The weapons are:

Hand grenades

Rifle grenades

Automatic-machine rifles

Rifles

The men who use these weapons are respectively known as:

Hand grenadiers

Rifle grenadiers

A. M. R. crews

Riflemen

As a consequence of the new weapons, a company is now organized on a basis of these specialists. A typical company is patterned like that of the French, the only difference being in the number of specialists. Each of the four platoons of a company contains the same number of specialists and each platoon is divided into four sections of specialists.

Organization of a Typical Company

One lieutenant (1st and 4th platoons under 1st lieutenants; 2nd and 3rd platoons under 2nd lieutenants).

One sergeant (second in command, assistant to platoon commander).

1st Section,	22	men
2nd Section,	12	men

3rd Section,	12	men
4th Section,	11	men
Total	57	men × 4 = 228 men
224 rifles		
64 pistols		
16 automatic rifles		

Platoon—1st Section—Hand and Rifle Grenadiers

- 1 Sergeant, pistol and rifle.
- 3 Corporals, pistol and rifle.
- 6 Privates, 1st class, all with rifles, two of them with pistols.
- 12 Privates.

—
—
men.
22

2nd and 3rd Sections—Riflemen

- 2 Corporals, rifle and pistol.
- 3 Privates, 1st class, rifle.
- 7 Privates, rifle.

—
—
12

4th Section—Automatic Riflemen

- 1 Sergeant, pistol and rifle.
- 1 Corporal, pistol and rifle.
- 3 Privates, 1st class (automatic rifle gunners, including 1 extra).
- 6 Privates, rifle.

Officers

- 1 Captain.
 - 3 First Lieutenants.
 - 2 Second Lieutenants.
-

6

The platoon is the self-contained unit with the proper proportion of all the different kinds of specialists for the assault. The reason for this is that the platoon is the largest unit that one leader can control in combat. The four platoons are alike and therefore interchangeable.

Although it is true that the specialists are particularly trained with their own weapons, all men of the company are first trained as riflemen. All specialists carry rifles, and if for any reason they cannot use their special weapon they immediately become riflemen. Also every man is trained in hand grenade throwing. This makes it possible to fill gaps in the specialists' ranks from the riflemen. Some men also, usually to the number of the A. M. R. crews, must be trained to operate the automatic-machine rifles.

The remaining supernumeraries of the company, such as mess and supply sergeants, mechanics, cooks, buglers, etc., do not march with the company in maneuver or combat formations. Some of the above supernumeraries will be found in the captain's headquarters, or tactical group, such as the liaison agents from the platoons, orderlies, buglers, and observers. Mess and supply sergeants, cooks, etc., remain with the regimental train in rear, with the kitchen behind the artillery positions. The tactical group of the captain lives in an adjoining dugout to that of their chief in the sector. In the fight these latter are used as messengers or observers.

II. Use of Specialists

The following table shows the evolution of infantry armament since the beginning of the war:

At Beginning of War

Rifle and bayonet	Nearly total personnel
Hand grenades	Nil
Rifle grenade guns	Nil
Automatic rifles	Nil

Present Time

Hand grenadiers per Co.	48
Rifle grenadiers per Co.	24
Automatic rifles per Co.	16

Outside the company is the 37 mm. gun in the battalion and trench mortars and engines, permanent equipment of the trenches.

In the Defensive

Hand Grenades. Hand grenades are used to produce a short-range defensive barrage at a distance of about 30 yards in front of the trenches. For this purpose one grenade thrower is required for from 10 to 16 yards of frontage.

They are also used to establish centers of resistance; to protect important features of the lines, such as salients, machine gun or automatic rifle posts, command posts, listening posts, etc.

They are used to defend a barricade in a communicating trench. In this latter case the grenades may be thrown from a special bombing post established for that purpose or from an adjoining trench.

Rifle Grenades. Rifle grenades are employed to establish a barrage at a greater range than that of the hand grenade, being used up to as great a distance as 300 yards. The great advantage of the rifle grenade barrage is

that it is under the control of the platoon leader and can be called into service immediately, which is not always true of the artillery barrage.

Several grenade rifles concentrated on a trench will stop hand grenade throwing from the same. Rifle grenades are usually shot from the cover trench immediately in rear of the fire trench.

Automatic-Machine Rifles. In the defensive the A.M.R's are usually employed to obtain flanking fire from the firing line, but may be used in the cover trench and sometimes in shallow holes out in front. In interior fighting they are used to cover stretches of communicating trenches against an enemy that has penetrated the first line.

Riflemen. The riflemen are placed in the intervals between the specialists in the firing line. They are also used as snipers and as sentinels. These sentinels are not to be confused with watchers, who may or may not carry rifles.

In the Offensive

Hand Grenades. In the offensive, hand grenades are used to reach the enemy under cover in trenches, etc., by high-angle throwing. It is an excellent weapon to clear out the trenches that the assaulting columns are advancing against. The H. G. and the R. G. break the resistance, so to speak, of the line that the attack is to capture and the assaulting troops are to occupy and hold. The places of the hand grenadiers in the assaulting column are seen in the diagram.

The hand grenade is the principal weapon in trench combat—that is, to obtain progress laterally and in depth in the trenches after a breach has been made at one point, as every foot of trench has to be fought for. Each section of trench, between traverses, is cleared of the enemy by throwing H. G's into it. It is then occupied and the next section cleared and occupied in a similar manner.

In a raid on the enemy's trenches hand grenades are used to clear up the particular stretch of trenches before the raiding party jumps into them. This will be taken up in detail later.

Rifle Grenades. The rifle grenade in the offensive is employed similarly to the hand grenade, to reach an enemy under cover, but at a greater

distance.

It is used to help reduce all nests of resistance that the assaulting infantry runs up against, and when the artillery barrage is not obtainable for that purpose.

They may be used as a barrage to cut off the retreat of a small group of the enemy that is being attacked in front.

Rifle grenades are very useful against counterattacks, especially before machine guns can be brought up and the defensive artillery barrage obtained.

The A. M. R. The A. M. R., with its great mobility and ease of control, makes it possible to carry this weapon in the assault and to be used immediately the wave is halted for any reason. In this connection, it can be instantly employed against a nest of resistance.

It is used in the assault to protect the flanks of the assaulting waves, especially when stopped.

As the A. M. R's are brought up in the first wave, they are immediately available against counterattacks upon the captured position. In a similar manner they are employed to hold ground gained until an artillery barrage can be obtained to support it and before the machine guns come up.

The Riflemen. The riflemen, with bayonets fixed, do the necessary fighting in the interior of the enemy's position. As was stated before, the H. G. and the R. G. break the enemy's resistance and the rifleman does the hand-to-hand fighting required to occupy the objective lines of trenches.

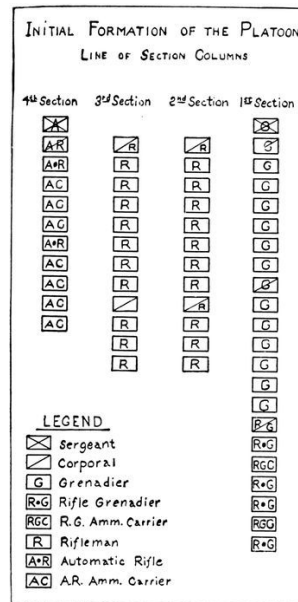
To make a little résumé, we may say that the best results will be obtained by the proper combination of all of these weapons. The low trajectory weapons (rifle, A. M. R., H. G., and the 37 mm. gun) concentrate on everything that shows itself above the trenches; while the high-angle missiles (H. G. and R. G.) seek out the enemy under cover. The riflemen complete the success of the specialists by rushing forward and occupying the enemy's position and fighting hand to hand for it if necessary.

III. Formations for Maneuvering this Resultant Company of Specialists

(Formation of the Platoon)

Initial Formation. The initial formation of the platoon in sections of specialists as per diagram is almost self-explanatory. Each group of specialists is in this formation under the control of its leader ready for any maneuver.

The platoon is the attacking unit because it contains all the specialists required for the assault. The frontage of an offensive is composed of a series of platoons, each with a definite objective, and each with reinforcing platoons in depth following.



All other formations, whether for combat or ordinary march maneuvers, are based upon this initial formation.

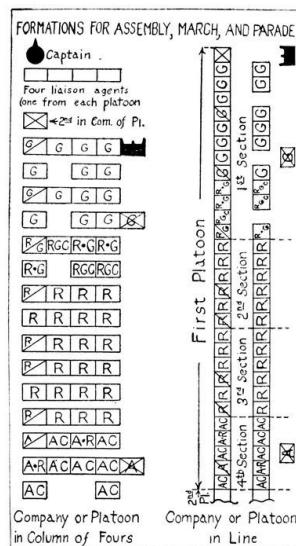
Close Order and March Formations. From the initial formation, we pass to the close order formations, used for parade, roll call, or the march.

(a) First we have the company in line, which is the habitual formation for assembly.

(b) From the company in line is formed a column of fours, for marching. The squads that are complete execute “squads right or left,” while the

remaining odd number of specialists in any section execute the easiest movement to bring them into column of fours, sometimes with blank files behind the rest of their section.

These formations are taken up under the conditions of open warfare, or, in the case of reinforcements, following up an attack.

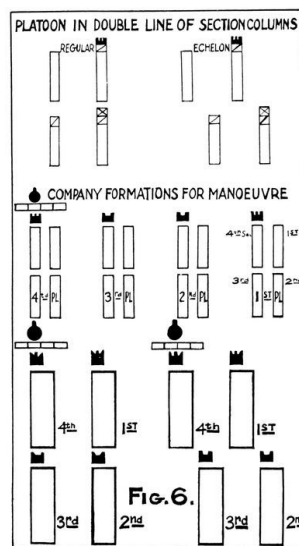


Skirmishers. This formation of skirmishers is not taken up for the assault, but under similar conditions as that of the approach march—that is, for open warfare and for the advance of the reinforcing platoons behind an attack.

To form the skirmish line, each section column simply deploys into line, each man going either right or left front into line from the regular formation of double line of section column.

If you inspect the figure, you will see that this deployment into skirmish line gives practically the type formation of the first reinforcing platoon in the assaulting column.

Assaulting Formations. In the attack the platoon forms and goes forward in waves. A wave is not one line of men, but is composed usually of two or three lines.

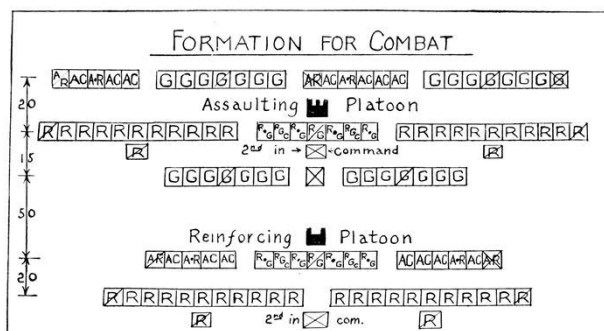


The composition of the lines of the wave depends upon the duties and functions of the different specialists during the attack.

Let us recapitulate the functions of the specialists in order to show the reasons for their positions in the assaulting wave.

The First Line of the Assaulting Wave. In this line are the grenadiers, whose duty it is to clear the enemy out of the objective trenches. Here also are the A. M. R's, employed to protect the flanks of the wave. When the wave is halted, the A. M. R's must be available for instant use against any nest of resistance and must therefore be in the first line. They are also used in the captured trench to withstand an enemy's counterattack, which may develop within five minutes after its capture.

The Second Line of the First Wave. We have here the rifle grenadiers who help the hand grenadiers break up the enemy's resistance and clear the trenches of the enemy. This is done by fire of rifle grenades from the second line. Here, also, we have all the riflemen of the platoon, whose function it is to follow up the success of the specialists and drive the surviving hostile men out of the trenches with the bayonet.



Third Line of the First Wave. In this line we have the “trench cleaners” or “moppers-up.” These are hand grenadiers furnished by the reinforcing platoon and their function is to clear the enemy out of the trenches, both laterally and in depth of the position. The assaulting column remains above ground and continues across the position to the objective line of trenches, which is usually the limiting objective of the attack. The “trench cleaners” picket all entrances and exits of dugouts to allow none of the enemy to come out after the waves have passed and fire into the backs of the assaulting troops.

These three lines compose the first wave, and we see that the specialists are placed in the lines in the order of their duties.

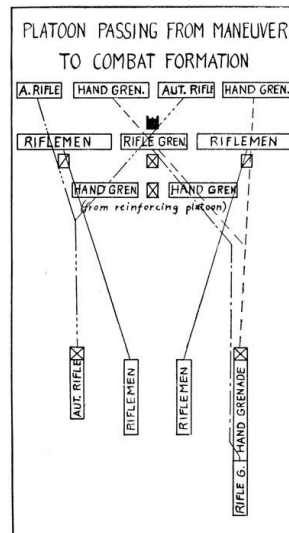
Company Drill Regulations Adopted at Fort Sheridan, Illinois, R.O.T.C., in Lieu of No Existing Official Regulations

Company Organization

1. For the purposes of drill and maneuver, the regiment will be organized as follows:

(a) One training company, as at present constituted, will form two platoons.

(b) The new organization contemplates four platoons to the company. When, for training purposes, a complete company is necessary or desirable, companies will be combined.



2. For a training company of two platoons, the 2nd in command (sergeant) of the first platoon is the right guide; the 2nd in command of the 2nd platoon is the left guide.

3. The platoons, in line or column, are arranged as at present.

4. The usual arrangement of a company for attack is to place two platoons in the 1st wave and two in the 2nd wave. The two platoons in the 1st wave are known as the assaulting platoons; the two in the 2nd wave, as the reinforcing platoons. The normal distance between the assaulting and reinforcing platoon is about 65 yards. Sometimes three platoons are placed in the 1st wave and one in the 2nd wave.

5. The first wave is followed, at a distance of 15 yards, by the hand grenadiers of the reinforcing platoon, who are sent forward by the commander of the reinforcing platoon to act as trench cleaners or moppers-up for the first wave. Fifty yards behind these is the remainder of the reinforcing platoon. There are no "moppers-up" for the reinforcing platoon.

6. The captain of a company has four liaison agents constantly with him, one being furnished from and for each platoon. They march immediately in

rear of the captain.

7. The post of the captain is between the 1st and 2nd waves.

Platoon Organization

1. The composition of a platoon is as follows:

1 Lieutenant (1st and 4th platoons under 1st Lieutenants).
(2nd and 3rd platoons under 2nd Lieutenants).

1 Sergeant (2nd in command, assistant to platoon leader).

First Section—(Grenadiers).

1 Sergeant—Section Sergeant.

2 Corporals and 12 Privates—Hand Grenadiers.

1 Corporal and 6 Privates—Rifle Grenadiers.

Second Section—(Riflemen).

2 Corporals and 10 Privates—Riflemen.

Third Section—(Riflemen).

2 Corporals and 10 Privates—Riflemen.

Fourth Section—(Automatic Rifle).

1 Sergeant—Charge of 1 gun crew.

1 Corporal—Charge of 1 gun crew.

3 Privates, 1st Class—Gunners—1 in charge of each gun and 1 in reserve.

6 Privates—Ammunition Carriers.

2. All platoons are alike in numbers, organization, and armament.

3. When the platoon is in column of squads or in line, the post of the platoon leader is as at present. When in line of section columns he is normally in front of the right center section, although he may be in front of some other section if he sees fit. In attack formation he is in the center of his platoon between his first and second lines.

4. When the platoon is in line, the post of the 2nd in command is in rear of the center of his platoon. In line of section columns he is in front of the 3rd section. In attack formation he is in rear of the fight center of the second line.

Platoon Movements

1.

1. The 1st Section (Grenadier Section) contains 15 rifle grenadiers (including the section sergeant) and 7 hand grenadiers. When in column of squads, the rifle grenadiers march as a squad, in rear of the hand grenadiers. The 1st Section constitutes 3 squads, each having the No. 3 in the rear rank as a blank file. The Section Sergeant marches abreast of the rear rank of the 2nd squad of his section. Line is formed by executing squads right or left.

2. The 2nd Section (Riflemen Section) contains 10 privates and 2 corporals. One corporal has charge of 7 men, the other of 3 men. When in column of squads, the section forms one complete squad and the front rank of another. Corporals are number four. To form line at the command "Squads left (right)," the complete squad executes "Squad left"; of the half squad the Nos. 3 and 4 execute "Twos left"; the Nos. 1 and 2 likewise execute "Twos left," thus placing themselves in rear of Nos. 3 and 4. To form column from line at the command "Squads right," the Nos. 3 and 4 of the incomplete squad move forward two paces, execute "Twos right," and place themselves in rear of Nos. 3 and 4 of the preceding squad; Nos. 1 and 2 move forward two paces, execute "Twos right," and place themselves in rear of Nos. 1 and 2 of the preceding squad.

3. The personnel, numbers and movements of the 3rd Section are identical with those of the 2nd.

4. The 4th Section (Automatic Rifle Section) contains 1 sergeant, 1 corporal, and 9 privates (total of 11). One complete squad is formed and 2 privates are left over who march in rear of Nos. 1 and 4 of the squad. The corporal is No. 4 of the front rank. No. 3, front and rear rank, are the gunners. The sergeant marches abreast of the rear rank of the squad in the line of file-closers; when line is formed, the two men left over form one additional file on the left (right) of their sections. In forming for attack formation, the No. 4 of the two men left over attaches himself to the front rank gun crew and goes with them; the No. 1, to the rear rank gun crew. The sergeant remains with his gun crew if the guns are separated, or assumes charge of both when together.

5. If the training company contains more men than are necessary to fill the platoons as indicated above, such additional men will be carried as riflemen and attached to the second and third sections of the platoons.

II

1. To form line of section columns from line.

The left file of each section moves straight to the front, followed by the file next on its right, and this in turn by the other files of the section. The section sergeants of the 1st and 4th sections place themselves at the head of their respective sections.

2. To form line of section columns in columns of twos or fours.

“Half-squads (squads) right front into line.”

3. To form line from line of section columns.

“Sections, right front into line.”

4. To form column of squads from line of section columns.

First form line of section columns in column of squads—then “Column of squads—1st (4th) section forward (column right or left).”

5. To form line of section columns from column of squads.

The left file of the leading section moves straight to the front, followed by the remaining files of the squad, then by the left file of the second squad, etc. The 2nd, 3rd, and 4th sections oblique to the right (left) in column of squads to section interval and, when in rear of their place, form column of files. The section may thus be marched in echelon, or line may be formed by halting the leading section, or the sections in rear may be double-timed to the line established by the leading section.

6. To form line of section columns to the flank, command:

“Sections, column left (right).”

“Line of section columns.”

7. Frequently, for convenience in maneuvering, the platoon leader may, when in line of section columns, wish to form only two columns from his four columns. In such event he will command:

“Double sections, 1st and 4th sections forward.”

One rifle section will follow the first section. The other rifle section, the 4th section. The rifle sections in such formation are always in rear of the specialist sections.

8. Intervals between sections or double sections are opened and closed by the command:

“To —— spaces extend (close).”

III

1. Attack formation is taken up from line of section columns *only*.

2. The attack formation is always in two lines. The two lines constitute a wave.

3. Hand grenadiers and automatic riflemen are always in the first line. The riflemen and rifle grenadiers are always in the second line in the assaulting platoon. Rifle grenadiers may be posted in the center of the second line, or on either flank, as the platoon leader directs. Similarly the automatic riflemen may be in the center, on either flank, or divided between center and flank of the first line, as the platoon leader may direct.

4. The 1st and 2nd lines are separated by a distance of 20 paces.

5. To pass from line of section columns to attack formation, the platoon leader commands:

“Form for attack at (*place*). Automatic rifles to center (left flank) (right flank) (both flanks) (left and center). Rifle grenadiers to center (left) (right).”

Only the automatic riflemen and grenadiers need be designated. The remaining men accommodate themselves to their movements.

At the above command the 2nd and 3rd sections stand fast until the 1st and 4th sections have cleared their front. They then move to their places led by the corporal at the head of their respective sections.

This movement will be executed at first in quick time; as men become more proficient, in double time and at a run.

IV

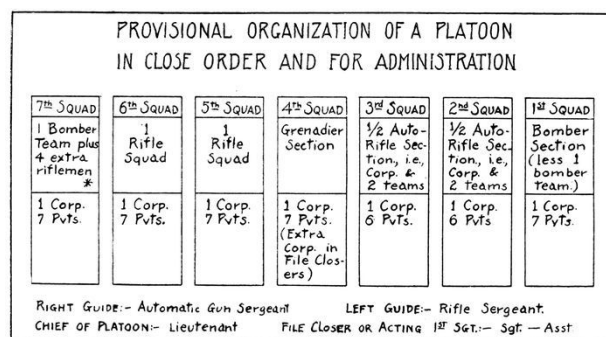
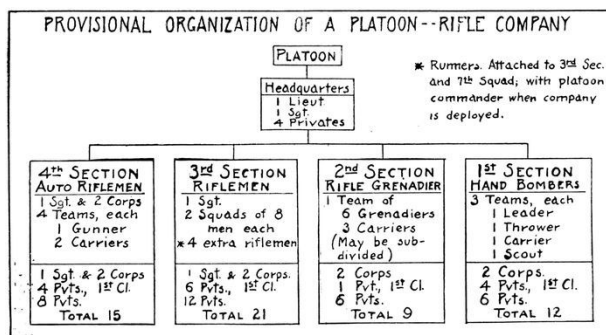
1. The arrangement of specialists in the reinforcing platoon differs somewhat from the assaulting platoon. The hand grenadiers of the reinforcing platoon are sent forward in two squads to follow in rear of the

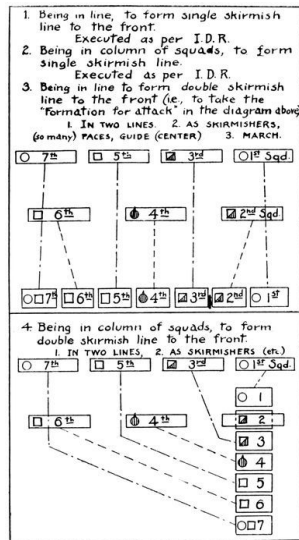
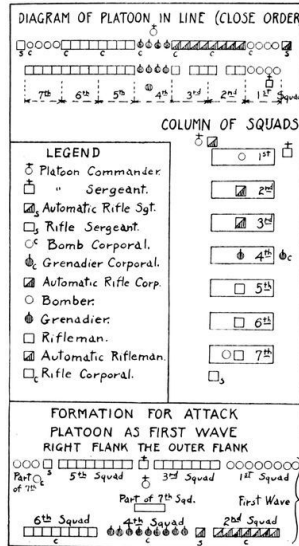
2nd line of the first wave at a distance of about 15 yards. The two squads are abreast of each other, the section sergeant between them.

2. The rifle grenadiers of the reinforcing platoon are then moved up into the 1st line.

3. The riflemen of the reinforcing platoon remain in the 2nd line.

4. The position of the platoon leader of the reinforcing platoon is in front of the center of the 1st line. The position of the 2nd in command is the same as in the assaulting platoon.

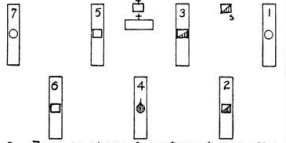




5. Being in line or column of squads, to deploy in line of squad columns in one or two lines. Same as 1 to 4 substituting Squad Columns for Skirmishers.

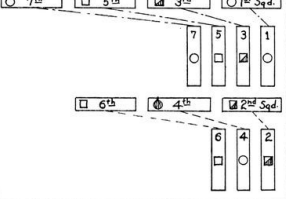
1. IN TWO LINES 2. SQUAD COLUMNS (so many) PACES, GUIDE (RIGHT) 3. MARCH.

← 150 Paces (more if possible) →



6. Being in above formation, to vary the intervals. 1. SQUAD COLUMNS, (so many) PACES, 2. GUIDE (CENTER) 3. MARCH.

7. Deployment from lines of squad columns



CHAPTER II

DEVELOPMENT OF A POSITION FROM AN OPEN WARFARE BATTLE

General Use of Fortifications. The first use of fortifications, whether hasty or permanent, is to give shelter against hostile fire. The more deliberate use of fortifications is to increase the holding or fighting power of troops by the proper employment of the natural defensive features of the ground and by artificial devices.

Successive Phases of the Organization

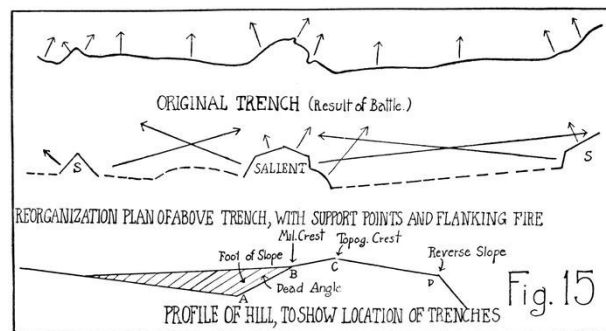
Change from Battle Lines to Trenches of a Position. During a battle, as we understand it in open warfare, as soon as the fighting lines are halted for any reason, every man immediately begins to dig in for shelter against the enemy's fire. If the halting of these fighting lines is more or less permanent these "skirmish holes" are connected, and we have the beginning of a rude trench. We must remember that in a battle troops are disposed in depth in a series of lines. All these lines dig for protection in a similar manner, giving the beginning of several lines of trenches. If the troops remain for any time in this position, it is necessary to have routes from the rear to the front to carry up ammunition and supplies to the different lines. These communication routes are also dug in, and we have the beginning of communicating trenches. Soon the men will begin to seek better protection against rain, cold, and overhead fire, and build themselves some kind of shelters. In a similar manner command posts and supply depots are established.

In time there is a complete skeleton of a series of lines of trenches outlining a position. If the mission of the troops is to remain and hold the ground, the necessary plan of organization for the final position will have to be based upon this skeleton. The purpose of this new organization, as we have noted, is to establish the position so that the ground can be held by fewer troops.

Active and Passive Elements of a Position, Especially Support Points. The first fire trench of a position is continuous, but it is not good practice to man the whole of this line, because this would take too many men and defeat the purpose of the fortifications. Secondly, a better defense can be established by garrisoning a series of strong points, or salients, from which can be obtained flanking fire. In the first line these are called support points. The trenches connecting these support points are retained as such, but are not usually manned. This curtain trench, so to speak, is retained for communicating purposes, and to deceive the enemy as to the real points of defense. A few watchers are left in it.

The first line, then, is a series of support points, well defended by men and special weapons, and at such distances apart as to support each other.

Division of Position. The garrison of such a support point in the first line is a company, and the captain is responsible for the holding of this ground. (The platoon, we must remember, is a tactical unit.) In dividing the first line into these active elements the size of each support point depends upon the importance of the ground or of the tactical situation.

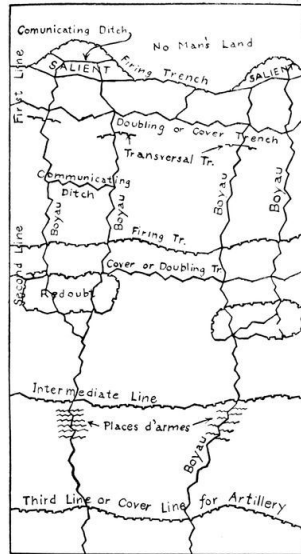


Similar natural strong points are also located in the second line. They will be fewer and farther apart in this line than in the first line, which is exposed to hostile attacks. These points in the second line are connected with the support points and together the group is called a center of resistance. The command of such a center of resistance, including one or two support points in the front line, falls to a battalion commander with his unit. For example, the area of a center of resistance under the command of a Bn. C. O. will include, say, two support points in the first line with a company in each, and the strong point in the second line with two companies.

In the third line, also, natural strong points are located and organized for defense. The command of this strong point, including, let us say, two centers of resistance, falls to a colonel with his regiment. Such an area is called a sector, or a sub-sector. If it is a sub-sector two such constitute a sector in command of a brigadier-general.

Different Lines of a Sector. We have now arrived automatically at the fact that each sector is composed of a series of lines in depth, each defended in strong points.

The first line is termed the line of resistance, as the first defense is made against the enemy's attacks in the support points of this line.



The second, or main line of resistance, is called the support line. In the strong points of the centers of resistance are the reserves and the headquarters of the battalion commanders.

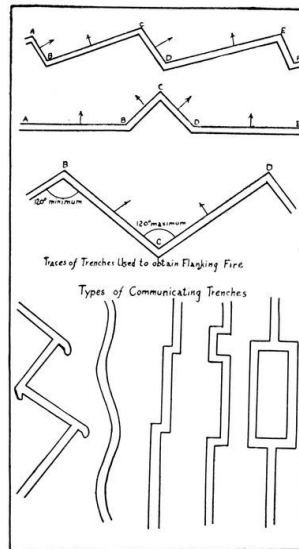
The third line is called the covering line of the artillery, because it defends the batteries distributed behind it. It also contains the reserve troops of the sector, a battalion usually holding each strong point. Here are established the headquarters of the colonel and the different central depots of the sector.

The above three lines constitute one position. There may be several positions in depth at distances of three miles or more apart. Usually there are two positions and sometimes three. It is said that the Germans have as many as six positions in depth in places. The first position only, of course, is permanently occupied.

Principles of the Organization

Advantages of the Continuous Trench. The continuous or curtain trench between support points must not be eliminated for several reasons. The trench will deceive the enemy as to the location of the elements of support points. If this trench were absent the enemy could easily locate these, and subject them to concentrated artillery fire. The position cannot be

hidden from airplane observation but the essential elements, such as emplacements, dugouts, etc., must be concealed.



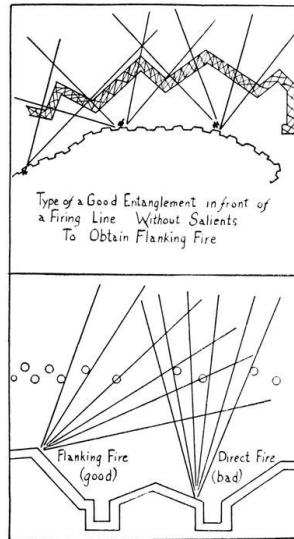
The curtain trench serves for communicating purposes for lateral movements of troops and for liaison between neighboring units.

The absence of the curtain trench would leave breaches in the line where in a general attack units could concentrate and break through. In such a general attack the curtain trench is occupied and defended by the reserves. Barbed wire, of course, protects the entire front.

These trenches also serve, if necessary, as cover for reserves acting as reinforcements to the adjacent support points, as there is only sufficient shelter in a support point for its own permanent garrison.

Lastly they may serve as trenches of departure in the offensive.

Importance of Flanking Fire. It is possible to have the support points at intervals because flanking fire from these will command all ground in front. The importance of flanking fire cannot be overestimated. Everything must be done to obtain it.



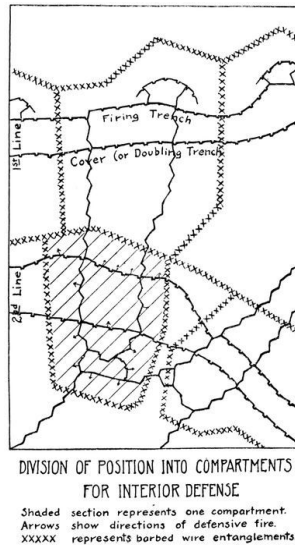
Sometimes it is impossible to obtain flanking fire because of lack of salients. If this is true the trace of the first line trench must be rectified to give flanking fire along the line as per diagram.

When neither of these two methods is possible small combat posts with machine guns are located in advance of the line to obtain flanking fire.

The machine gun is the best weapon for flanking fire. Several emplacements are constructed for each machine gun, but during the bombardment it is kept under cover in a dugout. When the attack develops the machine gun is quickly set up in the proper emplacement. If kept in a permanent emplacement the enemy will soon locate it and put it out of business by artillery fire.

If no emplacements are possible or if they are destroyed machine guns are operated on open ground or in shell holes.

Field of Fire. With proper flanking fire a line can be defended with a shorter field of fire than otherwise. A hundred yards of field of fire is sufficient with good flanking fire and accessory defenses, such as barbed wire, etc.



Location of Trenches. Lines of trenches which are the result of a battle are not always located in the most logical positions. They are the results of the exigencies of the battle. However, it is well to know the best locations for the trenches of a position under different conditions in order to place them there when possible.

In flat country, of course, it makes no difference where the trenches are located. On the profile of a hill, however, the question is where to locate the first and where to locate the second line. Let us consider Figure 15. Point A is out of the question as you have no view of the enemy. At B you can observe the enemy's line and he can observe yours. However, the enemy can shell you at this point and observe the results. At C the same conditions obtain as at B, except there is a dead angle directly in front. At D you cannot observe the enemy nor can the enemy observe your line. From these considerations we see that the proper location for the first line will be at B as you must be able to observe the enemy and all the ground in front. Your second line should be located at D where the enemy cannot observe and bombard your position. At D the line can be made as elaborate as you have time, men, and material, because you are more or less unmolested by the enemy's fire and observation.

The reverse slope is of the very greatest importance in organizing a position. Movements of troops and supplies can be easily accomplished here under cover. Deep dugouts can be constructed with the least work. But the greatest advantage lies during the bombardment. The enemy is not able

to observe the accuracy of his fire on the reverse slope so that the elements in this line are left more or less intact. For this reason the line on the reverse slope becomes the most important line of resistance against the enemy's attacks. The Germans habitually organize their reverse slopes very strongly.

Division of the Position into Defensive Compartments. We have already spoken of the sector in depth by successive lines. In the same way it is necessary to organize the positions for lateral defense. A support point may be captured by the enemy and from this ground he can launch a flanking attack on the adjacent part of the position. It is necessary that there should be an established defense against such an attack, and for this purpose each sector is divided into compartments, so to speak, with all-around defense.

Boyaux or communicating trenches on the flanks of these compartments are organized as firing lines with barbed wire belts running parallel and with machine guns disposed for flanking fire. We have then each sector cut up into compartments capable of independent resistance in all directions, if the surrounding compartments are captured by the enemy. Such a compartment also has the advantage of serving as a base for a counterattack against an adjoining one that has fallen into the hands of the enemy.

In a sector the responsibility of such lateral defenses falls especially upon the colonel.

Concealment. All of the essential elements of a position, emplacements, dugouts, etc., are more or less without value if they can be seen by the enemy and subjected to heavy artillery fire. Everything that is possible must be constructed without the observation of the enemy.

It is important to hide your works by making them fit in with the color scheme and shape of the surrounding ground. False or dummy trenches, emplacements, shelters, etc., are constructed to cause the enemy to waste his ammunition. The art of camouflage is also extensively employed for this purpose against both ground and aerial observation.

Elements of the Organization

Plan of Organization. We have seen that the first outline of the trenches left as a consequence of the battle must be modified in detail to obtain

better protection and organization. The complete plan of reorganization is laid down by the commander of the sector and is called the Plan of Organization.

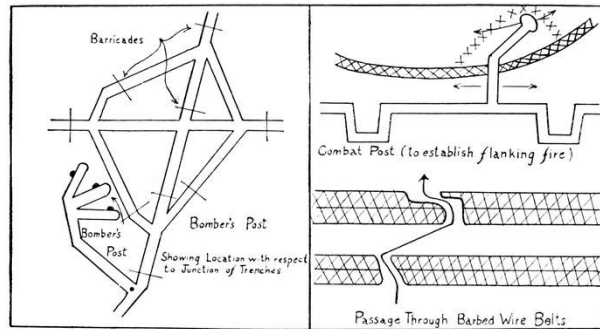
This plan is drawn up as soon as the position has a permanent garrison. The Plan of Organization must be faithfully followed by all the successive garrisons of the sector. After its adoption, even the commander of the sector himself cannot modify the plan without the consent of the general in charge.

The real value of the position depends upon the amount of continuous work that has been done in carrying out the original plan.

Outline of the First Line. As was seen before, the original trace of the first line was established as a consequence of the needs of the battle. In the organization plan, it is necessary to introduce several modifications in this trace. The commander responsible for this change should place himself in the situation of the enemy and consider an attack upon his own lines. From this consideration he will introduce such modifications as will make this task as difficult as possible.

The responsibility of making this modification in the trace of the first line does not devolve upon the captain in the first line. As a rule if it were left to him few changes would be made because of lack of initiative on his part, or because of fear that if the line were captured he would be blamed for having changed it. This task falls upon the battalion commander who is responsible for the center of resistance, with the consent of the sector commander. The reason for this is that the sector commander is the only one of these officers who knows the subsequent use of this particular sector, whether for defensive or offensive purposes. If for defensive purposes the line may be carried as close to the enemy's line as possible. If the sector is being organized for an offensive there must be enough distance between to keep his own line out of the zone of artillery dispersion when the target is the enemy's line.

Cover Trench. In general all fire trenches have a second trench behind, called the doubling or cover trench. The use of the cover trench is not primarily for defensive purposes. It has special uses.



The majority of the troops in the first line are kept in the cover trench in shelters located there. The first line is habitually manned by only a sufficient number of men to assure observation and security. The rest of the garrison remains in the shelters of the cover trench to get as much rest as possible. It is very important that comfortable shelters be located in the doubling trench. If troops are exposed to the elements and bombardment their value decreases very rapidly. If the shelling becomes too severe the men in the first line retire to the shelters of the cover trench until the attack develops.

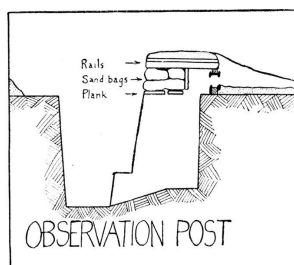
Another use of the cover trench in the defensive is to keep here a necessary number of troops for reinforcing the first line when necessary and for counterattack against the same if it is taken. This latter use establishes the distance between the two lines, which is from thirty to forty yards, so that hand grenades can be thrown from the cover trench into the first line.

Behind each support point there is usually a lateral communicating ditch called the circulating ditch which should not be confused with the cover trench.

Advance Posts. In front of the firing line certain advance posts are established. There are three kinds of these: listening, observation, and combat posts.

The *listening post* is not under the control of the commander of the support point but is in the service of the Intelligence Officer. It is furnished with a microphone for the purpose of picking up enemy's telephone messages in the hostile line.

The *observation post* belongs to the sector and is used for purposes of observation and security.



The *combat post* is established to obtain flanking fire along the barbed wire entanglements or it is used as a bombing post when in close proximity to the enemy's trenches.

Sometimes a series of combat posts are connected and a new trench established in front of the firing line. This is called the observation line. It is best to avoid this practice as there is always the question whether to hold or abandon this line when the enemy's attack develops. In either case there are grave disadvantages. If it is held you have a weak line. If you abandon it the enemy can use it for shelter.

Advance posts give excellent service if not too numerous. If there are many of them they are a temptation for hostile raids.

Dimensions of Trenches. These can be found in text-books on the subject. The general idea is that they are to be narrow and deep to prevent observation and to present a small target for high-angle fire.

Loopholes. Loopholes are used in the ordinary life of the trenches for observation and sniping purposes; but against the real attack of the enemy the firing is done over the parapet. If it is necessary to meet the enemy in hand to hand combat the men climb up on the parapet.

Traverses. Traverses are used to localize the effect of shell or grenade explosions, and to prevent enfilade fire.

They are sometimes used in long stretches of boyaux or communicating ditches as emplacements for A. M. R. to enfilade the same if the enemy penetrates the position.

Transversal Lines. Transversal lines are usually constructed between the first and the second lines and between the second and the third lines. They are short stretches of trenches parallel to the front for special purposes: such as disposing of machine guns and trench mortars in depth; bombing posts for rifle and hand grenadiers; depots and observation posts.

Command posts are usually located in such small transversal trenches, a little in rear of the lines they are in command of. For instance, the captain in charge of a support point might be located in a transversal just in rear of the cover trench.

Support Trenches. The second or support line is established at such a distance that the first line is covered by its fire if the latter is captured.

Also the second line should be placed so that it will be out of the zone of dispersion if the first line is the artillery target either of your own or of hostile fire. This places the distance between the two lines at from 150 to 400 yards. At this distance also rifle grenade fire can be used in conjunction with an artillery preparation for a counterattack against the first line.

The organization of the second line is the same as that of the first; in other words, the same principles and elements of defense are used: making employment of barbed wire, flanking fire, cover trenches, etc.

The second line is the main line of resistance and must be made as nearly impregnable as possible. For this purpose it must be thoroughly and completely organized.

In the support line are the reserves of the C. of R. and for their proper shelter they should have very deep dugouts, and all the requisites for the comfort and rest of the troops.

Redoubts. Behind the support line is a redoubt containing the command post of the battalion commander. It is the last strong point of the center of resistance and is organized for an all-around defense.

The line of redoubts is the last line of serious resistance. If the enemy penetrates this line a breach has been made in the position at this point.

Boyaux and Communicating Ditches. As we have said before trenches for communicating purposes must be dug both in depth and laterally. We shall call those from front to rear boyaux, and those running laterally communicating ditches.

The number of boyaux is not fixed as that of the different lines. There must be at least one between each support point and its center of resistance.

Usually boyaux are used for movements in both directions. Often certain boyaux are designated for entrance only and others for evacuation. The numbers of the latter are usually less than the former.

In an attack all boyaux are used for movements from rear to front.

All movements to the rear while reinforcements are being brought up must be executed in the open ground, exceptions being made in the case of wounded.

Boyaux are important for the movement of supplies and ammunition, but above all they are essential for reinforcements during battle. The greater part of the troops in a sector is held in reserve in the second and third lines. This disposition is only possible if you have good boyaux to the front lines.

All boyaux and communicating ditches are narrow and deep. Boyaux are wider between the second and third line than they are between the first and second. Evacuation boyaux are usually wider than others. The trace of a boyau must be such as to prevent enfilade fire.

At intervals of about ten yards turnouts large enough for a stretcher are constructed in the side of the boyaux for passage of troops in different directions. Boyaux are named and signs are put up similar to those for streets in a city. The same name of a boyau must be kept from the third lines to the first.

At branches of the main boyaux they are dug a little deeper to avoid mistakes of identification.

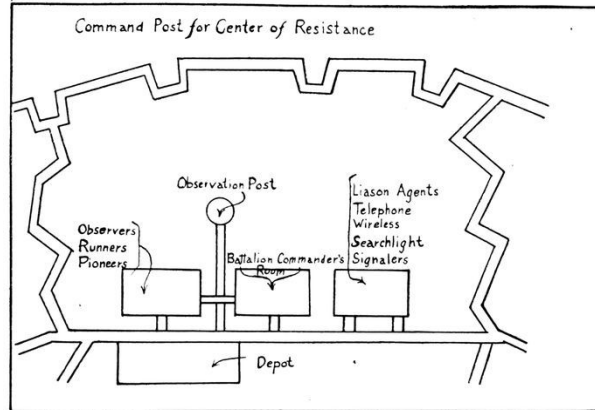
As was said before certain boyaux are organized for defense.

It may be said here that supplies carried up at night are usually brought along on top of the ground close to the boyaux. This gives freer movement and the carriers are close to the cover of the trench if needed.

Accessory Defense. Of all the accessory defenses such as barricades, abatis, trous-de-loup, etc., the most important by far is barbed wire, which is almost exclusively used for obstacles.

Barbed wire should be placed from twenty to fifty yards in front of the first line. At this distance shelling of the barbed wire belts will do very little damage to the first line.

The wire entanglements must be under fire of the first line.



The trace of the wire belt is not necessarily parallel to that of the firing line, but should be so placed as to obtain flanking fire along the wire belts. Good entanglements are constructed in double belts, the distance between the two belts being from ten to twenty yards, the farthest belt, of course, being constructed first.

Breaches must be left through the wire entanglements for the passage of patrols, friendly raiding parties, etc. Portable sections of barbed wire are used to close these passages when necessary.

All accessory defenses must be concealed so they cannot be easily observed and destroyed by the enemy.

Command Post. A command post, in the full sense of the word, contains the headquarters of the leader of a division of a sector.

The command post must first of all be a good observation post, after which the other requisites are installed. When the observation post has been constructed the following are built: a room for the headquarters staff to work and to sleep in, a telephone central, dugout for liaison group, and a depot.

The preceding discussion is a development of a position from battle conditions. On the other hand if you are unmolested by the enemy the order of organizing a position would be as follows:

- (1) Construct the emplacement to obtain flanking fire.
- (2) Construct barbed wire entanglements.
- (3) Observatory and dugouts.

(4) Trenches.

CHAPTER III

SECTOR AND TRENCH DUTIES

I. Plan of Defense

Definition. The Plan of Defense contains all the different duties necessary to make a proper defense of the position. The execution of these duties is the execution of the Plan of Defense, which follows logically and depends directly upon the Plan of Organization.

Principles of the Plan of Defense. (a) The most important principle of organization of a position is the **disposition of troops in depth**. It is necessary to have plenty of support and reserve units disposed one behind the other. Only one-sixth to one-third of a garrison in the support point of the first line remains in the trenches for the purpose of observation and security. The rest remain in the rear as a reserve for reinforcements or counterattacks.

(b) Each trench, each nest of resistance, each area of the ground, must have a responsible leader. **The garrison of any such area must never leave the ground allotted to its defense under any conditions of battle.** The defense must be carried out by this garrison on the lines originally laid out. A leader is not allowed to fall back on troops in his rear or to take up a better position. The troops in depth are placed there for the purpose of reinforcements for the lines in front.

(c) **Each part of a position that is lost to the enemy must be immediately retaken by a counterattack** carried out by troops held in reserve. The first line is retaken by reserve troops in the support line, and similarly, if the support line is lost, it must be retaken by reserve troops held in the third line. If all three lines are captured, an attempt must be made to retake the third line by the reserve troops of the sector.

Form of Plan of Defense. In each division of the position from front to rear there must be a Plan of Defense. This plan always contains the five following elements:

1. Directions of the probable attacks.
2. Defense of the first line, or line of resistance.

The outline of the first line.

Location and mission of the troops in the first line.

Employment of machine guns.

3. Counterattacks.

Units which are used for the purpose.

Directions or trenches to be used in the counterattack.

Depots of grenades and sandbags for the counterattack.

4. Artillery support.

Composition of this artillery.

Establishing of the front of barrage.

Infantry-artillery liaison.

5. Liaisons and supply during the hostile attack.

Details of Trench Duty

Guard Duties (Security). Security is obtained by the use of sentinels and patrols.

Sentinels must not be confused with observers, watchers, or snipers. Observers belong to the information service of the sector and are not on duty at night. Snipers operate only in the daytime. The watchers of the different lines give the alarm in case of an attack, and are on duty both day and night. Sentinels are posted in the first line or in advanced posts. Their work and number will depend upon the proximity of the enemy, the tactical situation, and above all upon the conditions of the wire entanglements out in front.

At night, one sentinel is posted, say, for example, every 10 yards in the fire trench of the first line. During the day, only enough sentinels are posted so that all parts of "No Man's Land" can be observed. This will allow a maximum number of men to rest.

In the firing line, the sentinels work in three reliefs. When not on guard duty, they sleep. The remainder of the garrison in the cover trench, however, are available for work, for patrols, and to furnish supply parties. Sentinels

should have their rifles resting on the parapet close at hand, ready to fire at a moment's notice.

By day, the sentinels use a periscope or observe through a loophole. This practice is, however, strictly prohibited at night, when a sentinel must observe over the parapet. As little challenging as possible is done, and then only in a very low tone.

Reconnoitering Patrols are the best means for security against attack. Every night the commanders of the support points or the centers of resistance send out patrols in front of the firing line. The strength of a patrol must be sufficient to carry back a wounded man, *i. e.*, the number should be five, six, or more. The success of a patrol depends entirely upon the leader. He must be clever, daring, and observant, and the rest of the patrol is simply his bodyguard. He himself must observe the objective of the mission of the patrol.

Patrols should never be sent out without definite orders as to their mission. Such may be, for example, a reconnaissance of the enemy's lines, to discover if the enemy has cut passages in his wire belt preparatory to an attack, to examine their own wire entanglements, etc.

The route of the patrol must be exactly determined; also the hour of departure and return designated. Word is passed along the line of sentinels that a patrol is out in front with the necessary information about the same. The preparation for a night patrol must be made during daylight.

Officer of the Watch. In the firing line the company commander arranges a roster of his officers for duty as officer of the watch. Also in each platoon the platoon commander will detail a N. C. O. of the watch. The officer of the watch is responsible for the duties of observation and security, and is especially responsible for the giving of the alarm in case of an attack and consequently for the calling of the artillery barrage by use of rockets, signals, etc.

Standing to Arms. It is of the greatest importance that the troops in the first line are thoroughly trained in taking their positions when the alarm is given as quickly as possible, without confusion. The alarm is often given for practice in order to ascertain the amount of time it will take the men to reach their places in the firing line.

The command "Stand to" is usually given one hour before daylight and one hour before dark. At this time the company commanders arrange for

inspection of arms, ammunition, and equipment by the platoon commanders.

Observers' Duties. Every effort should be made to obtain information of the enemy, his trenches, and his wire entanglements. Any alteration in the enemy's lines must immediately be reported. The importance of forwarding such information must be impressed upon all ranks. This duty is carried out by observers.

All men of the company are used as sentinels in the first line or as watchers in other lines, but not all are observers. Men who are to act as observers receive special training. A man to become an observer must have a special capacity for this task.

Each platoon should have at least six such observers, enough for proper reliefs.

The observers are placed:

1. In each company the observers are distributed in a number of observation posts properly camouflaged so as not to be discovered by the enemy. These posts may be in advance of the firing line, in the firing line, or at any point within the position that has a wide view of the front. The observers employ periscopes or field glasses. Everything of note that is observed is recorded and from these data the captain makes out his daily observation report.

2. Also in the sector there are 2 or 3 observatories from which the information is turned in directly to the colonel. These posts must be properly concealed and protected. In each one of these we find the following:

- (a) Regulations posted that give the personnel, tour of duty, sector of observation, direction to special points of interest, and certain precautionary measures.

- (b) Specially prepared blank forms upon which to record information gained by observation.

- (c) Field glasses, maps and sketches, and panoramic views of the ground in front, both visible and invisible from the observation posts.

- (d) A telephone connecting the post with the nearest command post (center of resistance or sector).

The observers in these posts are directly under command of the intelligence officers of the regiment and are of a special personnel. They must be thoroughly familiar with the regulations of the posts, maps, panorama, etc.

The intelligence officer collects and coordinates all information gathered in the sector. Every morning he turns in a consolidated report made up from the information in the captains' daily reports and from the records of the sector observatories. (In order to avoid confusion of the different daily reports these information records are kept on two different colored blanks, red and blue alternating each day.)

The report of the intelligence officer is signed by the colonel and sent on to the brigade headquarters. From the data of his complete report the intelligence officer very often must verify certain information by the use of patrols. For this purpose he makes a request upon the colonel for special patrols.

Information is gathered from both airplanes and ground observatories. From ground observatories it is attempted to obtain the following data;

1. To ascertain the intentions of the enemy.
2. To properly prosecute attrition of the enemy's works.

Attrition. By the word attrition in fighting is meant the gradual destruction of the enemy's morale and strength, and demolition, little by little, of the enemy's position. Trench warfare is not a rest or simply a time for guard duty. It is a phase of the battle in which you must do as much damage to the enemy as possible.

This work will also keep up the offensive spirit. For this purpose different means are used:

(a) Sniping: Each company in the first line uses its best shots as snipers. A sniper must be a good observer and he must depend upon his keen eye for evidences of the enemy, such as smoke of cigarette or the glint of a bayonet, giving proof of the presence of an enemy. Especially during hostile shelling will a too curious enemy come to his loopholes to observe the effect of the same. This is the sniper's chance.

In each center of resistance the snipers are placed under the command of a sniping officer. He makes the rounds of the snipers' posts once or twice a day, and submits a daily report containing known casualties of the enemy,

location of the number of snipers' posts, telescope rifles, fixed rifles, and rifle batteries.

(b) **Infantry Weapons:** Besides sniping, all the other infantry weapons are used to assist attrition of the enemy. Hand and rifle grenades are thrown on special points of the hostile line. A battery of hand or rifle grenadiers properly disposed can very often carry out a concentrated surprise fire on an enemy's salient or advanced post. Automatic and machine guns are always used for this purpose and especially at night. They are trained on supply routes, road junctions, etc., and by use of sudden bursts of fire at irregular intervals they will compel the enemy's supply parties to continue their transportation in the boyaux instead of on the open ground.

The use of indirect machine gun fire is very depressing to the enemy. Such judicious employment of machine guns is sometimes more effective than severe artillery shelling.

The 37 mm. gun, which is also an infantry weapon, is used principally against hostile machine gun emplacements. The location of this gun must be constantly changed.

(c) **Use of Artillery Weapons:** Besides furnishing barrage fire against hostile attacks, the artillery can be called upon for surprise fire on working parties, harassing fire during the night against supply parties and their routes, and also for reply fire against hostile shelling.

Trench mortars and other engines keep the enemy as uncomfortable as possible at all times.

(d) **Fighting Patrols and Raids:** Fighting patrols and raids are used with the express purpose of causing loss and damage to the enemy. This is accomplished by engaging the enemy's patrol and working parties or by raiding his saps, listening posts, or trenches. Such operations must not, however, degenerate into frivolous local fighting, causing loss of men and waste of artillery ammunition.

Work. Daily trench work consists of (1) maintenance and (2) new construction.

(a) **Maintenance:** The daily work of maintaining the elements of the position is considerable, and its importance often escapes those responsible for the same. They are:

Repair of parapets, boyaux, and shelters damaged by bombardment, water, or frost.

Replacing or completing accessory defenses, revetments, ammunition recesses, emplacement of machine guns, observation posts, depots, etc.

Cleaning and Draining of Trenches and Boyaux: The garrison of each area is responsible for their own maintenance work, which is done usually during daylight hours. Sometimes during severe bombardments this repair and maintenance work becomes too great for the garrison and workmen have to be called in to help.

(b) New Construction: New elements of the position are constructed in carrying out the Plan of Organization. These are usually carried out at night. This work is carried out by troops in the doubling trench, the support line, or detachments of workmen from the reserve of the sector. There may also be special parties from the headquarters detachment of pioneers or of the engineer battalion of the brigade.

For all these special works the personnel for whom they are built must be present during the construction to furnish necessary information and for inspection. A machine gun crew helps in building its emplacement; the intelligence officer directs the construction of his observatories. The medical officer present during the work on the medical aid station; the telephone officer superintends the establishment of his telephone exchange, etc.

The battalion commander is responsible for all the work done in his center of resistance, and in conjunction with the sector leader draws up a time-table for the work.

All this work must be continued by each succeeding relief. When a new unit takes its turn in the trenches the orders and plans of new works are turned over to its commander.

Arms, Equipment, and Ammunition. Full equipment will always be worn in the firing trench, except the haversack, water bottle, and entrenching tool. In the support and reserve trenches the equipment to be worn is subject to the will of the sector commander. Every man will always carry a gas mask in any part of the position and even as far to the rear as ten miles.

The men in the firing line and in the doubling trenches will at all times be in possession of their rifles and bayonets. The rifles of men in the support and reserve trenches may be kept in protected racks in the same shelter as the men.

All rifles must be kept scrupulously clean at all times.

Ammunition is protected from bad weather by being placed in boxes located in the sides of the trenches.

Ten to twenty hand grenades are placed in a sandbag. These are usually kept at the depot and carried in these quantities for the supply of the fire trench in combat. One or two of such sandbags are kept at a bombing post.

Special care must be taken to keep rockets in good condition. Supplies are kept in the fire line (platoon leaders' posts, observations) and at points in the rear (in the observatories or different command posts or in the observation posts of the information service).

Depots and Supply. (a) Depots:—There should be one general depot in each area located near the command post of the chief of area. However, in order to facilitate work in the first line the platoon leaders may establish near their post a small depot of tools and material being used for the time being. The importance of the depot depends upon the area it is to supply. For example, the depot of the garrison of a support point in the first line would contain:

20,000 cartridges

1,000 hand grenades

400 rifle grenades

100 each—signal and illuminating rockets

60 to 70 gas masks

100 shovels

100 picks

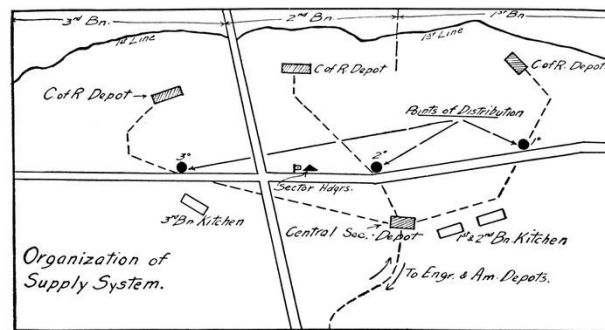
Sandbags, planks, barbed wire, etc.

The depot at the center of resistance contains the same articles but in greater number, also telephone wire, blankets, and three days' rations of food.

The main depot of the position, however, is the one near the command post of the leader of the sector. It is divided into three parts:

1. Ammunition and rocket signals (Ordnance Service).
2. Material of every sort (Engineer Service).
3. Food and clothing (Quartermaster Service).

The different services in the rear of the position maintain the supplies of the sector depot. The amount of ammunition to be carried is fixed by orders from the general and this amount must be strictly maintained.



(b) Requests for Supplies: Each morning the chiefs of areas send in a request for material required for the next night's work. All these requests in a sector are grouped by a staff officer under the heads of the three different services. Along with each request must go the information showing necessity for same. Emergency requests are made by telephone. The officer making request for material must be on hand to receive and check the same upon delivery.

(c) Supply Parties: The supply of the depot is usually made with special detachments taken from the support or reserve companies. If it is necessary to take men from the front trenches for this purpose, no more than 10% can be taken away from these trenches at the same time. Complete units will be used as supply parties with their leader if possible. Upon each battalion in a C. of R. and upon each regimental reserve falls part of the responsibility of distributing supplies in the rear of the sector. The unloading or rendezvous points are disposed along what is called the supply line, usually on a road, path, ravine, behind a hill, etc. Supply parties sent out at night to these points make as little noise as possible. They return along a well-picketed

path, usually running along one of the central boyaux. Transportation is thus carried in the open ground as far as possible, usually to the support line.

When proper protection is possible, pack trains are used to carry supplies right through the distributing point and up to the depots of the sector.

(d) Cooking: The battalion kitchens are located behind the sector in a protected emplacement. Steps must be taken to ensure as little smoke as possible being seen from them. At night each platoon sends a ration party to the kitchen.

When the regiment has rolling kitchens, these are brought up part of the way towards the sector where they meet the ration party. Arrangements should be made to always have, if possible, some hot soup or drink available for the men between midnight and 4 a. m.

(a) Activities of the Troops: Sector duties include a considerable amount of work, and are not only useful from a tactical point of view, but are also indispensable for the moral and physical welfare of the troops. Men without occupation in the trenches stagnate, grow slack, and think only of the time when their relief will arrive. Such troops lose their aggressiveness, so that when the time for the offensive arrives they have no confidence and cannot deliver the proper blow.

Good practice to obtain fruitful results is the employment of time as follows: In each company the captain prepares a daily schedule of duties in which he allots the different services of the strong point: hours of “standing to” and inspection, of sending ration and supply parties, time of rest, of cleaning arms, equipment, and clothing, time of trench work, etc. This schedule is communicated daily to each platoon at a regular hour.

(b) Discipline in the Trenches:—Life in the trenches is not especially dangerous when all precautions are observed. In the presence of an alert enemy, every faulty move of disposition receives its punishment, such as the improper concealment of a movement, poorly hidden trench work, or useless noise in the firing line. Silence is one of the essential rules of the trench. Silence in a sector in which troops are properly disciplined enables the chief to rapidly transmit his orders and the men to accomplish same without delay.

Daily Reports:—In each sector, from the captain up, the different commanders must furnish a daily report. These are co-ordinated in the

headquarters of the sector commander, and afterwards sent to the division headquarters.

(a) Telephone Reports:—

Night report. 5 a. m. to 7 a. m.

Day report. 3 p. m. to 5 p. m.

These reports contain the main events of the time covered. Similar messages are sent at any time important events occur, such as heavy shelling, important information of the enemy, activities in the air, etc.

(b) Written Reports:—The daily morning written reports contain the following paragraphs.

1. General aspect of the sector.

2. Important events of the day.

(The events must be reported in detail, for example: A report of shelling must contain the details as to the area bombarded, the caliber of the gun used, and the time and direction of same, etc.)

3. Casualties (wounded, killed, and missing).

4. Consumption of ammunition.

5. Requests for miscellaneous equipment and material.

This report is signed by the chiefs of the area responsible for the information in the report. Along with this official report is added a second written report with the following paragraphs:

1. Request for materials for the night.

2. Observation and information reports (to the Intelligence Officer).

3. Report of all the work done in the sector, including time schedule of the center of resistance.

Use of Infantry and Artillery Weapons, and Liaison

Machine Guns. The tactical use and location of machine guns in the sector depends upon the following two points:

(a) Machine gun emplacements must be thoroughly concealed and during the bombardment the M. G. itself is kept under cover in a dugout. Previous to an attack of the enemy, his artillery searches the position for the

destruction of the machine guns and their emplacements, because they are the most powerful weapons of the defense.

If invisible casemates can be constructed this practice is good. The best method is, however, to prepare several emplacements, properly camouflaged, with a deep dugout near at hand for the protection of the M. G. and its crew. This shelter must be located at such a point that the M. G. can be quickly brought out to any one of its emplacements when the attack develops. These emplacements usually are simple pits located in front of the trench and connected with the dugout by a gallery. When emplacements are destroyed by hostile artillery fire, or when they are not possible, machine guns are pushed into shell holes or operated on the open ground. Shell holes out in front give the best service as emplacements for a daring M. G. crew. The French success at Verdun depended to a great extent upon the judicious use of shell holes for M. G. flanking fire.

(b) Machine guns are not only located in the first line, but are disposed throughout the interior of the position. The hostile artillery may entirely destroy the first line of defenses, but the proper disposition of machine guns for flanking fire in the interior lines will prevent the enemy's progress after penetrating the position. Machine guns are so located in the position as to cover all the space between the different lines with flanking fire. Usually three pickets define the sector to be covered by any one M. G., one at the M. G. and two limiting its sector of fire.

In a sector, $\frac{1}{3}$ of the machine guns are located in, or near the first line, $\frac{1}{3}$ is disposed of in the support line or in the redoubt. The last $\frac{1}{3}$ is assigned to the third line. In each line machine guns are under the command of the chief of the area.

As a general rule, in the first line, automatic-machine rifles, because of their mobility, are used instead of machine guns.

Standing orders are, that M. G. crews will never surrender, but fight to the last man. Often the tenacity of a M. G. crew has permitted the retaking of an area otherwise lost.

Trench Mortars. In each sector there are located a certain number of trench mortars. These are efficient weapons, but must be used with care, because they will nearly always draw concentrated artillery fire.

They are used for the destruction of special points in the enemy's lines that the artillery cannot reach.

They are employed in the preparation for an offensive against the enemy's first line, either for an attack or for a raid. They are used for reply fire against enemy's trench mortars.

They participate in establishing barrages.

The efficiency of trench mortars cannot be overestimated, but at the same time, great precaution must be taken in their use and concealment. The personnel required to serve the light trench mortars are taken from the garrison of the sector and belong to a special headquarters unit, a platoon of bombardiers.

The large trench mortars are commanded by an artillery officer who keeps himself constantly in liaison with the commander of the center of resistance. He consults the latter on the location of his mortars and especially upon their use. He also reports to the commander of the center of resistance all orders he receives from his artillery chief. This artillery officer always works in co-operation with the chief of his area, who in turn gives him all necessary information and help. Especially does the mortar commander require help for the supply of ammunition.

Artillery Support in a Sector

Composition of the Artillery Support. The light artillery of this sector is not directly under the command of the Division Commander. The Commanding Officer of the light artillery, usually a Colonel, in co-operation with the chief of the sector, divides it into different parts, assigning one to each of the centers of resistance. This artillery is called support artillery. As we shall see later this practice is usually the same in the offensive.

The amount of artillery allotted as support depends upon the tactical situation, in other words, the activity of the enemy.

In the Verdun and Champagne offensives the disposition of light artillery was the same in principle as that for the defensive, that is to say, one troop of artillery consisting of 3 batteries was assigned to each infantry battalion.

The general use of this artillery, is:

(a) To protect the infantry with barrage fire.

(b) Prosecution of destruction of the enemy's personnel and works.

Tactical Liaison between Infantry and Orderly. To carry out its functions, the artillery support must be in perfect liaison with the infantry, and only when this is done is co-operation of the two arms possible. This liaison is obtained by the following five means:

(a) Organization of Leadership:—For the full co-operation of the infantry and artillery, the latter is not kept entirely independent, but is divided and assigned to infantry unit. Each chief of sector and each chief of center of resistance knows what artillery must support him if he needs it. The commander of the artillery support is not exactly under the command of the infantry chief of area, but he is at the latter's disposal for certain calls for support fire. The proper co-operation of these two officers will give efficient results.

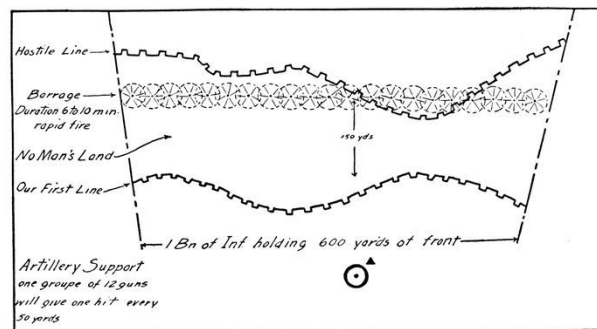
(b) Relation between Artillery and Infantry Commanders:—The best of relations must be established between the artillery and infantry commanders. Usually, it is not possible to locate the commanding posts of the two officers close together. Each has his area or unit to command and for that purpose must be located for the proper control of the same. The support artillery commander, however, should make frequent visits to the infantry leader and his area. He should also send his subordinate artillery officers into the infantry area to become acquainted with the first line, the conditions in "No Man's Land," the enemy's lines, and to gather all the information possible from the infantry officers in the trenches. Everything should be done to foster good relations between the infantry and their support artillery and this is best done by frequent visits of the artillery officers to the infantry trenches to gather information for artillery data.

(c) Communication of Orders:—It is necessary that the orders received by the infantry commander be communicated in whole or in part to the artillery commander, and vice versa. For instance, if the chief of area receives orders to send out a fighting patrol or a raid, this information is sent to the artillery commander, who, in turn, sends back his plan of action. Similarly, when the artillery commander expects to carry out special fires not called for by the infantry, such as destructive shelling, harassing fires, etc., he should first notify his infantry chief of area.

Similarly, any useful information picked up, either by the infantry or artillery, must be communicated to the other. The daily report of the chief of

sector and the observation report of the Intelligence Officer are sent direct to the commander of the artillery support. Valuable information obtained at the artillery observation posts is sent to the infantry.

(d) Means of Communication:—The principal of the organization of the different means of communication is such as to assure the best liaison between the two arms. Telephone lines, searchlight signals, runners, etc., are established directly between a unit of infantry and its artillery support. This is to avoid loss of time and useless interruption between leaders.



(e) Detachments for Liaison and Observation:—For more efficient liaison between the artillery and the infantry, usually a certain number of artillerymen are sent from the group of the artillery Support to the infantry in their area. Usually a special detachment of artillerymen is sent. This detachment is made up of one officer, chief of the detachment, who is sent to the chief of sector, one N. C. O. to each chief of a center of resistance, the rest as telephone operators and artillery orderlies. This disposition of artillerymen in the infantry area is obligatory for an attack, but it is so practical that it is now employed in the sector organization.

The task of these artillery officers or N. C. O. is:—

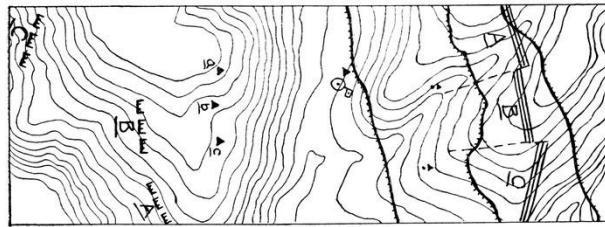
(1) To control all means of communication with the batteries of the artillery support.

(2) To transmit to the artillery the different calls for fire, and in technical terms to give the proper data for the laying of the pieces on the target, and the kind of fire required.

Use of Artillery Support. Different kinds of fire are executed by the artillery support. The most important is the barrage.

Barrage. Purpose:—The purpose of a barrage is to stop a hostile attack before it can get started, or to hinder enemies' reinforcements or reserve troops from coming forward.

Form:—The barrage is a curtain of fire let down in “No Man’s Land” as close in front of your first line as possible without loss to your own garrison. This will be about 150 or 200 yards to the front, between the two lines of trenches.



LOCATION OF ARTILLERY FOR BARRAGE

Front of the Barrage:—The barrage has real value only if it is impassible. This will be if the density of the fire is so great that the danger zones of the explosions of the individual shells overlap each other. The allotting of one troop of artillery, 12 guns, for an infantry battalion is quite sufficient for this purpose if the front of the battalion does not exceed the average length of from 400 to 600 yards.

Duration:—The barrage is a burst of fire which lasts from six to ten minutes. At the beginning the fire is very rapid and towards the end becomes slower. For instance, for the first six minutes the fire may be six rounds per gun and in the last four minutes, say, four rounds per gun. Successive barrages can be carried out depending upon the needs of the infantry.

Calling for the Barrage:—Barrage fire, for the best results, must open up at the moment the hostile attack begins. To accomplish this the call for the barrage must be made by an observer in the first line. For this reason every officer in the first line has the authority to call for the barrage. The barrage is the only fire that can be called for automatically by signal.

The kind and color of the rocket signifies the type of barrage required and the front it is to cover. In order to avoid mistakes this rocket signal is repeated from some point farther to the rear, usually at the observatories of

the command post of the center of resistance, or of the sector. Rocket signals are repeated until the barrage fire opens.

Confirmation of the barrage is given by telephone, if possible. However, at the first signal rocket the artillery is required to open up the barrage fire as quickly as possible without waiting for confirmation. Barrage fire is at the disposal of the infantry and cannot be refused by the commander of the artillery, whether it is justified or not.

Other Kinds of Artillery Fires. The commanders of the center of resistance or of the sector may call on the artillery support for other fires besides the barrage, such as:

Reprisal fire on the enemy's lines.

Reply fire to hostile shelling.

Surprise fire on supply attachments or working parties.

Destruction fire on special points.

All these fires are called for by telephone, but do not follow automatically like the barrage fire. The artillery commander, in such case, can use a certain amount of discretion whether these fires should be carried out or not. Last of all there is the counter-preparation fire, which is a strong shelling of the enemy's position. It is carried out as a reply to a methodical bombardment of the enemy preliminary to his offensive. It is the best means to prevent a hostile attack. If this fire is well executed the enemy's assault is "killed in the egg" as the French expression has it.

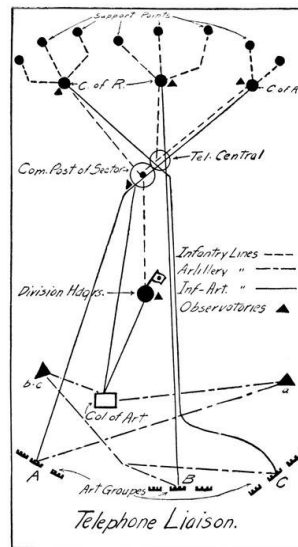
Liaison in the Sector.

(a) Telephone: There are two different systems of telephones.

First, there is the system connecting the infantry command posts, support points, centers of resistance, sector, and division headquarters, with the command posts of the artillery protecting the different infantry areas. These are direct lines between the command posts and the groups of support artillery. Also, this system connects the command post with observatories, depots, medical aid stations, kitchens, etc.

Secondly, there is the system connecting the chief of artillery with his subordinate commanders, the artillery observation posts, and the artillery dumps. In this same system the artillery command posts are again connected with the infantry units which they are to support. Special lines

are run by the artillery to the officer and N. C. O. of the artillery attachment of liaison in the infantry areas.



Consequently, the liaison between the artillery and the infantry is doubly provided for in these two systems.

The lines of connection between the infantry and the artillery must be separate from those within the infantry area and those between the artillery commander and his subordinate units.

Confidential information should never be telephoned except in cipher. Conventional designations are employed to call the different posts. Strict regulations are maintained as to who should use the telephone and for what purposes.

The weak point of a telephone system is that it is often broken by bombardment or earth-slides. This can be partially remedied by constructing as few direct lines as possible, by burying the wires in the bottom of the boyaux, and by executing repairs as quickly as possible.

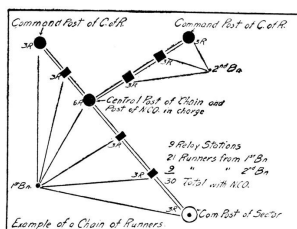
(b) Wireless and Ground Telegraphy: The battalions in the first line have ground telegraph apparatus connecting them with the regimental headquarters, and these, in turn, can communicate with the general headquarters by wireless. This means of communication is only employed when the telephone cannot be used.

(c) Signalling: It is difficult to establish signal communication within the sector without being seen by the enemy. However, at night with flashlights it is possible, especially in hilly country, to communicate from front to rear within the position. The receiving post should be able to certify received messages. When practicable, communications should be established between the command post of the center of resistance and the sector command posts, and from the latter post to the artillery. This method of liaison is organized as a duplicating means or as a temporary substitute for the telephone.

(d) Rockets: Signal rockets are used to communicate with the rear. Different forms and colors of rockets are used to signify different things. As we have seen, the most important use of the rocket is to signal for barrage fire. It is often necessary to relay a rocket signal from a second position in the rear to make absolutely sure that it reaches the artillery support.

Signal rockets are also used to communicate with aéroplanes, but this means is usually only employed during an attack.

(e) Runners: The liaison established with runners is best of all, from the point of view that it usually never fails. During heavy shelling it is the only sure means of communication.



The ordinary messengers or orderlies are not used as runners. A chain of trained runners is used, located in a little post, shelters, or shell holes; Each post contains three runners, their distance being from 150 to 300 yards apart. Each runner is selected from the unit located near the next post in the chain, that is, the one to which he is to run. Often the divisional cavalry furnish these runners.

The chain of runners is under the command of a N. C. O. who is responsible for its efficiency. He has an important task on his hands. He must assure himself that every runner knows, not only his two neighboring

posts, but also the general direction of the chain. He must change disabled runners and fill gaps in his chain. It becomes his duty to instill into the runners the importance of their duty.

(f) **Carrier Pigeons:** Carrier pigeons are the last means of communication. They have been found to give excellent service during heavy shelling or gas attacks. They do not fly very easily at night.

Each command post of a center of resistance in the first line has a pigeon post. These consist of two reliefs of men, four pigeons, and the necessary equipment and food. The pigeons themselves have to be relieved every few days. The care given to the birds at the command posts must not be as good as that at the dovecot in the rear, so that when they are released they will return immediately to the latter place, from where the message is delivered to its proper destination.

Instructions for Sentinel, Watcher, and Observer

Sentinel. A sentinel is a soldier placed in the first line or in an advanced post for the purpose of security.

The duty of the sentinel is to watch and guard against hostile attack and to warn the chief of the platoon of every action of the enemy in “No Man’s Land.”

At night, one sentinel is posted about every ten yards in the firing trench. During the day, only a sufficient number are posted to assure that all parts of the enemy’s line and “No Man’s Land” are observed.

Sentinels are relieved every two hours except under bad weather conditions when the length of the tour of duty is reduced. Every sentinel is regularly posted by a non-commissioned officer who explains to him his duties and ascertains that both the sentinel and his relief are aware of the position of the platoon commanders and of the sentinels, on either side.

At night, the sentinels must observe over the parapet. As little challenging as possible is done and then only in a very low voice. If the sentinel receives no answer to his challenge, he shoots, thus giving the alarm. During the day, the sentinel uses a periscope or observes through a loophole.

The sentinel always has the bayonet fixed on his rifle, which is loaded and ready for use. He must be on the alert for every noise, and for this reason he is not allowed to wear any ear covering.

Sentinels must be informed of the sending out of patrols, the approximate hour of departure and return, and the general direction that the patrol is to follow.

Watcher. Each platoon in the trenches, from the first line to the rear, must have some watchers, whose duty it is to give the alarm both day and night against attack. These watchers do not observe the enemy as do the sentinels. They walk in the trench in which the platoon is placed, especially at the entrance of the dugouts or shelters occupied by this platoon. When they hear a signal of alarm given by a sentinel or by a messenger, they wake the platoon as quickly as possible. In general, they give to the chief of the platoon all the information that they have obtained during the time of their duty. When an officer or private from another sector enters the trench, it is the duty of the watchers to ascertain the purpose of his visit. Watchers receive special orders to give the alarm against a gas attack, and must sound the alarm gongs and bells for this purpose.

The watcher may have his rifle, without bayonet, resting on the side of the trench at the entrance of a dugout.

Observer. The observers are specially trained men whose duty is to observe the hostile position and to gain all the information that they can obtain of the enemy's activities.

Each man in the company can be used as a sentinel or as a watcher, but not all as observers. To become an observer, a man must have a special capacity for this task. Each platoon should have at least six such observers of proper qualification; good eyesight and hearing, patient, and a good shot. They are relieved every two hours. Observation duty is carried out in daylight.

The observers are of two kinds:

(1) In each company the platoon observers are distributed in a number of observation posts especially chosen for this purpose. These posts are located in the support point of the company which may be in advance of the firing line, in the firing line, or in any point within the position that has a wide view of the front. These observers employ periscopes or field glasses.

Everything of note that is observed is recorded and from this data the captain makes out his daily observation report.

(2) Also, in the sector, there are special observatories belonging to the headquarters of the regiment, under the command of the intelligence officer of the sector. There are usually two or three observatories, from which the information is transmitted directly to the colonel. These observatories are furnished with field glasses, maps, and telephones.

In general, all observers are required to obtain all possible information of the enemy, to record all modifications of the hostile position in its works and wire entanglements, all the movements that they can observe, all working parties that they can discover. Also, they receive special orders for the cases they must report to the commander, as for instance: shelling of the position by artillery or trench mortars, gas attacks, signal rockets fired in the front line, activities of the aviation services, whether of friend or enemy.

CHAPTER IV

THE RELIEF

I. Definitions and Principles

There are two kinds of reliefs, General and Interior reliefs.

A **General Relief** is one where a large unit, such as a division or an army corps, is relieved from its position on the front.

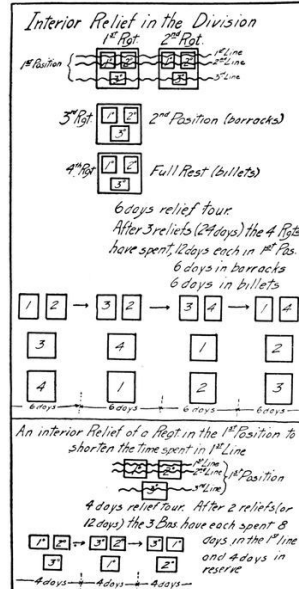
An **Interior Relief** is one where a small unit, such as a company, battalion, or regiment, is relieved from its area by another unit of the same division within the position.

A general relief takes place for the following reasons:

In order to send a unit far to the rear to good billets for complete rest and perhaps for recruiting.

To withdraw and gather together the units of a corps, for the purpose of training for an offensive.

To permit the strategical movement of large units along the entire front.



General reliefs should not be made any oftener than necessary as they hinder methodical and continuous organization of the defense, observation of the enemy, and the preparation of contemplated offensives. It is for the express purpose of decreasing the number of general reliefs that troops are

disposed in depth in a position so that continuous defense of the sector will be assured by means of successive interior reliefs. As an example of interior reliefs, we will consider a division holding a part of the front with two regiments disposed in the first position, one regiment near the second position in billets, and the fourth still farther to the rear in complete rest. These regiments, by a system of interior reliefs, will rotate to equalize the tours of duty in the first position. Similarly the battalions of the regiments in the first position will rotate to give equal divisions of time to each in the first lines. Assuming that a period of six to eight days spent in the front line is a fair average, and considering the fact that troops are disposed in depth throughout the three lines, different combinations of reliefs are possible. The work of making out the schedule of reliefs falls upon a division of the general headquarters.

The sector period is variable with the activity of the enemy and conditions of life in the trenches. During the German offensive at Verdun the sector period was four days, while in Lorraine during that time the same size unit could remain in the trenches for three months without necessity of relief.

II. Preliminary Dispositions for the Relief

Time to Prepare the Relief. Usually the order for the relief is issued from the general headquarters to the regiment forty-eight hours previous to its execution. Sometimes, however, for tactical reasons, such time cannot be allowed and the regiment may have to go immediately into the trenches. In such case, the different operations of the relief, which are taken up in this chapter, are shortened but must not be carelessly executed. In the present condition of trench warfare the proper execution of a relief is of the greatest importance.

Preparation. A relief is executed at night. Thirty-six hours before the relief is made the men of the relieving units proceed to clean and get in shape their arms, equipment, and clothing. When the relief is not for a special offensive operation, the troops go into the trenches with full equipment. All non-regulation articles are left behind with the regimental train, in care of the non-commissioned officer left with it. Also, the files, books, and official papers of the unit are left here. The regimental train establishes itself in rear of the sector of its regiment.

Besides this material preparation, the officers of the relieving companies are responsible for the morale of their troops. Before going into the trenches there may be some apprehension on the part of the troops, which must be dispelled by the officers. The great factor of depression is the mystery of what is not known of the sector about to be entered. Platoon leaders and company officers try to clear up this difficulty by telling their men what is known of the sector they are going to hold, and otherwise speaking words of encouragement.

Reconnaissance. Prior to taking over a sector, a preliminary reconnaissance is made by a certain number of officers and orderlies. They are:

For the Regiment—The colonel and part of his headquarters staff (adjutant, intelligence officer, telephone officer, engineer officer, medical officer, supply officer).

For the Battalion—The battalion commander and his adjutant.

For the Companies—The company commander and one officer for each company. Machine gun company commander and one machine gun officer. The commander of the 37 mm. gun platoon.

Messengers and telephone men at the discretion of the adjutant and the company commanders.

The above reconnaissance parties will go into the trenches on the morning of the day preceding the night relief. They must start early enough to permit a certain number of officers of these parties to return to the billets with all details and information necessary for the movement of the relief into the sector. The adjutant and the lieutenants of each company remain in the trenches until the relief arrives. During the intervening time they must acquaint themselves with all details of the sector necessary for the execution of its defense.

This preliminary reconnaissance affords an opportunity to the different leaders and chiefs of services to note the particular things of interest to each.

For the Colonel or Battalion Commander—The general organization and defense of their area.

For the Adjutants—The works under construction, precautions against gas attacks, etc.

For Officers of the Regimental Headquarters Co.—Information as follows:

Information Service—Activity of the enemy, microphones, observation posts and observatories, carrier pigeon posts, signal rockets.

Telegraph Service—Telephone systems, flashlight posts, wireless or ground telegraphy installations.

Engineer and Ordnance Service—Depots and supply (material and ammunition), special engineer works, trench mortar emplacements.

Medical Service—Dressing stations, evacuation of the wounded, hygiene (latrines, etc.).

Supply Officer—Emplacements of the kitchens, food supply.

For the Company Commander—Location of the platoons (number of men holding the first line and distribution of same; platoons in the cover trench and support line). Shelter accommodations, alarm signals, artillery

support (limits of barrage fronts and barrage calls). Counter attacks (troops and directions for counter attacks). Information of the enemy (observation posts of the support point). Supply (engineer dumps, kitchens, ration supply parties, water).

For the Machine Gun Company Commander—Distribution of machine guns and emplacements, limits of machine gun fire sectors, emergency and special uses of M. G.

When such a complete reconnaissance cannot be made, only commanders of units precede their troops and make a short reconnaissance of one or two hours.

Relief Orders

(1) **Regimental Orders.** Usually the general order for the relief of a sector is settled on in conference by the colonels of the relieving and relieved regiments. This order is sent to the battalion commanders, if possible, before the reconnaissance.

The order of relief contains the following paragraphs:

Day and hour of the relief, designating the relieving and relieved units.

The name, limits, and division of the sector with designations of the neighboring units.

Information of the artillery support, its composition and emplacements.

Orders for the reconnaissance, its composition, and the hour when it must be finished.

Orders for the movement of the relief; march of the regiment from the initial point to the point of dispersion where the battalions branch off to their particular areas. If the regiment is transported in motor trucks, points and orders for entraining and detraining. Hours of departure of each battalion from the dispersion point and special itineraries for each.

Movement of the relieved troops. Assembly points of these battalions.

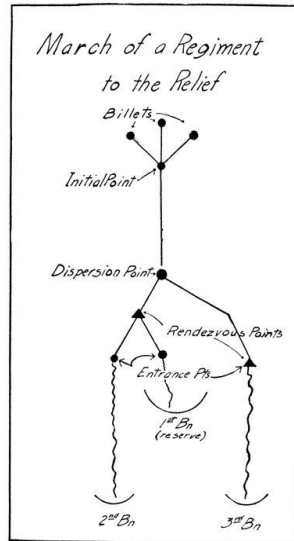
Movement of the regimental trains, relief of the kitchens, supply measures for furnishing of provisions, orders for the sanitary personnel.

Hour at which the new colonel takes command of the sector.

(2) **Battalion Orders.** All details for the relief are fixed in the battalion relief order issued by the battalion commander. This order is sent out immediately upon his return from the preliminary reconnaissance and in accordance with the regimental order.

The battalion relief order contains the following points:

Hour of leaving the dispersion point.



Order of march of the units (the companies march in the order in which they are to relieve the units in the sector, from right to left in the first line, and similarly in the support line).

Itinerary to the entrance of the trenches.

Meeting of the guides, hour and point of rendezvous, function of the guides of the relieved battalions.

Movement of headquarters, and rendezvous of the messengers that each company sends to headquarters to establish liaison.

Order of supply.

Additional details (result of the reconnaissance).

Departure. Before departure, each company is inspected by its company commander, who must see that each man has his canteen full, his rations for the day, a full supply of cartridges, the magazine of his rifle loaded (no cartridges in the chamber), and his gas mask in good condition. In winter the relieving troops will find a supply of blankets and trench boots in the

sector. Usually each man should carry an extra supply of cartridges above that laid down by regulation, and two sandbags. There should be a certain number of flashlights and candles to the company (each corporal carrying a minimum of two candles). Just previous to departure or entraining the battalion commander himself makes a general inspection of his unit.

III. Movement of the Relief

March to the Rendezvous Point of the Guides. Usually the regiment marches, but it is sometimes transported in motor trucks, up to a point in the rear of the sector, called the point of dispersion, from which the different battalions branch off successively to go to their respective areas. This practice avoids the crossing of units. A short meal may be had by the regiment at the point of dispersion. The distance of this point behind the sector and the time of arrival of the relieving unit must be carefully calculated so that the relief may be finished early enough to permit the unit relieved to get out of sight of the enemy before daylight.

When the battalions leave the rendezvous point, each company sends to the battalion headquarters two messengers for the purpose of liaison. The units now follow the prescribed itineraries up to the rendezvous point where the guides are met.

Guides. During the reconnaissance, the company to be relieved details certain men to act as guides for the incoming company. An average of two guides is furnished for each platoon, one for each company headquarters, and one for each battalion headquarters. These guides await the arrival of the different units at a certain point called the rendezvous point of the guides. Usually this point is at the entrance of the boyau. As several guides are usually grouped at the same rendezvous point, they must be alert to meet the unit to which they are assigned. A non-commissioned officer of the retiring battalion is in charge of this group and he is responsible that each guide finds his unit. Each guide, of course, must know the best and safest route by which to conduct the unit to the position it must occupy.

March in the Boyaux. From the rendezvous point, the companies continue their march in the boyaux which are assigned to them. One guide leads each platoon while the second guide brings up its rear. The leading guide must inform the platoon commander of the different points of interest that are passed, such as each line of the sector, regimental or battalion command posts, depots, water points, etc.

The march in the narrow boyau of an unknown sector, in the darkness, and with full equipment, is difficult and depressing. Consequently, the rate

of march must be slow. When the head of the column meets an obstacle, a warning word is passed to the rear along the single file. This rule is above all useful to prevent accident to telephone wires.

The march must be executed without noise. Orders are given in low tones. Smoking or the use of flashlights is prohibited. No disciplinary measure is too severe that will prevent the enemy from discovering the relief. It is absolutely prohibited to talk over the telephone concerning a relief otherwise than in cipher.

IV. Occupation of the Position

Details of the Relief. If the two units have the same number of men, the relieving of one unit by the other is simple enough. But often the relief is complicated by the fact that either the extent of front or the number of effectives is different. In such a case, the officers of the two units must settle, during the reconnaissance, upon the necessary modifications for an effective and expeditious relief.

When the company arrives at the command post of the captain, the platoons are relieved from right to left in the first line, then in the cover trench in the same order. If there is a platoon in the support line this is next relieved.

In the first line, the relief of the platoons is executed in two parts. First of all, the sentinels, observers, watchers, men in the listening posts, and the N. C. O. of the watch are relieved. When this is done, the remaining men of the platoon are relieved. The men of the old platoon occupy their places at "Stand To." The relieving platoon files in and steps up on the firing step. At the command "Pass" which is given quietly, the old and new platoons change places.

Each retiring leader, of whatever command he may be, hands over his orders and information to the corresponding leader who relieves him. For example, one platoon leader will turn over to the other all information concerning guard duty, defense of the line, condition of the barbed wire entanglements, patrolling, and shelter accommodations. This must be executed rapidly but with precision.

The platoon relieved assembles at the entrance to the boyau, and leaves the trenches under the chief of platoon. This is not done, however, until the platoon leader has reported to his captain, by means of his guides, the execution of the relief. The old captain fixes a point for the assembling of his platoons, but this point must be far enough to the rear to avoid blocking of the boyaux and out of reach of hostile grenade and trench mortar fire.

In the meantime, at the command post of the support point, the new captain acquaints himself with all information and orders pertaining to the support point from the old captain. When this is finished he telephones the

command post of the center of resistance the completion of the relief, and asks if his predecessor may retire. The relieved company then takes up the march, following the prescribed route up to the assembly point of the battalion.

It may be stated here that the relieved units must leave their trenches in as clean and sanitary a condition as possible. The work of clearing up the shelters and latrines must be thoroughly done before the arrival of the new units. Depots must be left in good condition and contain the amount of supplies called for by regulations. The platoon commander is responsible that his men do not forget, tools and cooking utensils in the trenches.

Duties after the Relief. In each support point, the new captain immediately establishes liaison with the command posts of the neighboring units to right and left. Communication between the captain and his battalion commander is obtained by means of the two messengers detailed to the battalion for that purpose and who now return to their company.

When the battalion commander has received the information from all his captains that the relief of his area is finished, he reports the same by telephone to his colonel. A confirmation of this report will be made the next morning in his daily written report which will cover the following points:

- The general condition of the relief.

- Hour of completion of the relief.

- Casualties during the relief.

- Living conditions in the trenches.

- Requests for tactical modifications (new dispositions of the garrison).

The relief finished, the battalion commander must also establish his liaison laterally and to the front and rear, but especially with his artillery support, by means of the artillery N. C. O. detailed to his area for that purpose.

Finally, the battalion commander investigates and perfects the organization of the food supply.

In the sector headquarters, the new sector commander will find on file all records, reports, orders, and detailed information concerning all the elements of defense of the sector. The sector file will contain the following documents:

Plan of organization.

Plan of defense.

Files of information concerning:

Machine guns.

Trench mortars.

Artillery support (table of barrages and other fires).

Liaison (telephones, runners, signalling, etc.).

Supply and evacuation.

Measures to combat gas attacks.

Besides these are the files of the different reports:

Sector daily reports (kept by the adjutant).

Intelligence reports (kept by the intelligence officer).

Construction reports (kept by the engineer officer).

Maps, sketches, and aëroplane photographs (kept by int. officer).

Inventory of the sector depot (kept by engineer officer).

Any officer of the sector, as for example a machine gun officer, has access to this file to more thoroughly acquaint himself for the execution of his duties.

CHAPTER V

DEFENSE OF A POSITION

Different Kinds of Hostile Attacks

Surprise Attacks. These may be raids or local attacks. A raid is usually for the purpose of taking prisoners; and a local attack, to capture a part of the line. These operations may be carried out without a preliminary bombardment, but are usually preceded by a short intense bombardment to destroy wire entanglements.

Attacks in Force. Such an attack may be made on a larger part of the line, as a particular sector, or on a considerable length of front in a general offensive. This kind of attack is, of course, preceded by a bombardment.

Special Attacks. Special attacks are made with gas, liquid fire, and mine methods. These will be taken up in detail in a later chapter.

Surprise Attacks

Precautions Against Surprise. In considering precautions against surprise attacks, it is not a question of combating patrols or enemy's reconnaissance parties whose missions are to search out information of your intentions and situation. It is a question of raids and little attacks on your lines. Against such hostile operations the defense of the sector depends upon the following precautions.

Maintenance of the Barbed Wire Entanglements. A surprise attack, to be successful, must have besides surprise the element of swiftness. The enemy must make a dash across "No Man's Land" up to your first line. This, however, is impossible, if your barbed wire entanglements are intact. Consequently each captain is responsible for the maintenance of the wire belt in front of his line.

It is the duty of the observation service to discover breaches or defects in your wire. To this end, the captain sends out at night patrols to ascertain the condition of his entanglements. If necessary, small detachments are sent out at night to make repairs. If this work is considerable, he calls upon the battalion commander for special working parties to assist. The captain, in

his daily requisitions for material, provides for the supply of wire, chevaux-de-frise, etc., that he may need.

Service of Guard and Observation. One of the most important duties of the observers in the first line, and of the sentinels and patrols, is to locate breaches that the enemy has made in your wire, or to discover hostile parties in the act of tampering with the same.

Breaches or passages in your wire may be made by such means as bangalores and petards and detonators. The explosion of such an arrangement is a sufficient signal of alarm. These are usually poor methods.

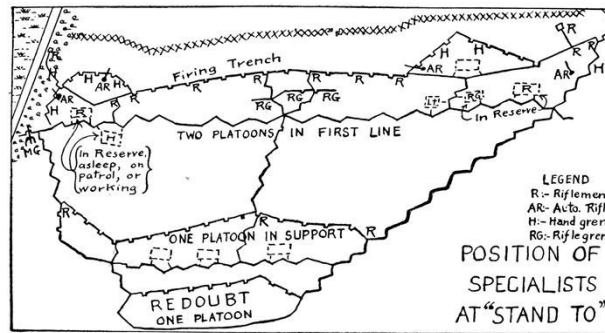
Passages may be made by special hostile patrols using wire cutters. These wire cutting patrols may precede the attack. Alert sentinels or your own hourly patrols, that crawl along your wire belt, should discover such operations.

The usual method, however, is to destroy barbed wire by artillery fire. The object of hostile shelling of your position is not always easily detected. The enemy may carry out a general artillery fire on your position during the day, for example, just before dusk. At the same time he will concentrate certain batteries for a methodical destruction of parts of your entanglement. For this reason it is the mission of the first patrol, sent out at dusk, to ascertain the condition of the barbed wire belt. The report of this patrol may carry important indications of the intentions of the enemy.

Sometimes a hostile surprise attack is preceded only by a short and unexpected bombardment. Registering shots for this artillery fire are made during the day. These registering shots, however, are fired short of the wire in “No Man’s Land” to deceive the observers that they are for the purpose of barrage control. In this case, the only means to baffle the attack is to have diligent sentinels to give immediate alarm, and a well-trained garrison to take its place in the fire trenches quickly.

“Stand To” Exercises. A well-trained garrison, accustomed to “stand to” quickly, without noise and confusion, cannot be easily surprised. When the call to arms is given by a sentinel, this call is repeated by all watchers in the successive lines of the support point. The officer of the watch runs immediately to the place of call. All the men of the garrison spring out of their niches and shelters, and proceed quickly, without orders, to their appointed places of combat. This execution must be automatic, and only a frequent repetition of the “stand to” exercise can give a unit this result.

Besides these exercises, as we know, the garrison executes “stand to” one hour before dusk and one hour before daylight. Deep dugouts are prohibited in the first line because they may become traps in which the defenders are taken prisoners, and they materially lengthen the time for the garrison to take its place for combat.



Only in case of an expected attack and when the garrison is ready for it, is the alarm signal given by such special means as bells, whistles, or rockets. This is exceptional. The only practical alarm signal that will awaken the garrison any time during the night is the cry “to arms” passed along and taken up by everyone.

Rifle, A. M. R., and grenade fire from the first line will usually stop a surprise attack. But since it is impossible to ascertain the strength of the attacking troops, the defensive artillery barrage is called for. Similarly, the reserve troops located in the cover and support trenches may be used for the purposes of reinforcement and counterattacks, depending upon the success of the enemy’s attack.

The captain of a support point must not only keep up the offensive spirit of his garrison by all the means of trench fighting, but he must organize and maintain a vigilant guard and patrol service for security.

Attack in Force

Attacks in force are carried out to capture an important position, or there may be a series of such objectives along a front in a general offensive. On the other hand, the object of the defense is not only to repulse the enemy but also to inflict upon him as great a loss as possible.

The phases of a great attack are: the preparation, of which the final and most important element is the bombardment; the assault of the first line; and the exploitation of success by fighting in the interior of the position. The defending troops must combat each one of these phases by the most adequate measures. We shall take up these measures in detail.

I. During the Preparation of the Hostile Attack

Revealing Symptoms of the Attack. The preparation of the hostile attack reveals itself by different preparations:

(1) The activity of the enemy:

His infantry will send out more numerous and aggressive patrols and reconnaissance parties.

His artillery will increase its daily destructive fire.

It will multiply its number of registering shots, establish barrage fires in the interior of your position, and begin to concentrate on communication routes.

His air service will take on unusual activities.

(2) Vigilant observation of roads and territory behind the enemy's lines may reveal convoys carrying troops and supplies, the assembling of troops in woods, etc. (smoke from their kitchens), accumulation of materials in depots (often accompanied by explosions in artillery dumps).

(3) The observation service may discover new hostile works, such as the contraction of parallels of departure out in front of his first line. There may be also successive parallels behind the nearest jumping-off trench.

(4) Prisoners and deserters, who become more numerous because of the approaching offensive, may give valuable information concerning the coming attack.

The task of discovering the enemy's preparation for an attack is incumbent upon all ranks, but above all, this duty devolves particularly upon the information service, balloon and air service, and artillery observers. In the sector, the intelligence officer must perfect his system of observation to the highest degree. The information in the daily reports is carefully sorted and tested.

Preventive Measures before the Attack. During this time, each chief of area takes all preventive measures possible against the coming attack. The attack is an expected event for which all must be prepared. The preparation simply consists in perfecting and carrying out of the original plan of defense.

Perfecting the Organization of the Position. New lines are created. The number of continuous lines between strong points and successive lines for defense in depth are increased.

Old lines, which are enfiladed by the enemy or which do not give flanking fire, are rectified or abandoned for new ones. The number of boyaux and communicating ditches are multiplied to facilitate the bringing up of reinforcements.

New accessories of defense are constructed and the old ones strengthened. This is especially done in the interior of the position with respect to parallel and perpendicular barbed wire belts to divide the position into compartments capable of all-round defense.

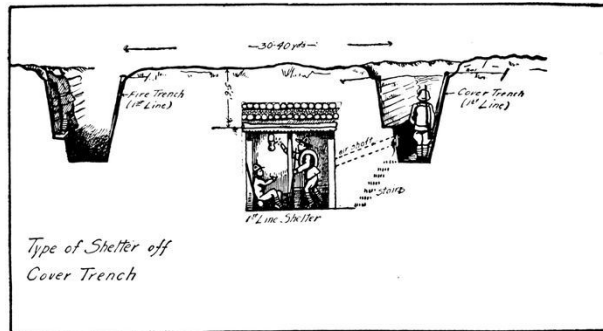
Additional false works are created, to waste the enemy's artillery fire. The art of camouflage is extensively used during this phase. New shelter accommodations and dugouts are constructed for the better protection of the garrison during the preliminary bombardment.

Increase of the Control and Discipline of the Defense. The guard and observation services are reinforced. Sentinels, observers, and patrols are held to a stricter account of their duties.

Machine gun leaders are particular to perfect the disposition of their guns in depth of the position for interior fighting. The sector for each gun emplacement is carefully verified. Dugouts serving as shelters for guns during the bombardment are strengthened.

The support artillery must constantly verify the settings of their barrages. Numerous exercises are carried out for the call of the barrage with rockets, so as to keep the artillery constantly on the alert.

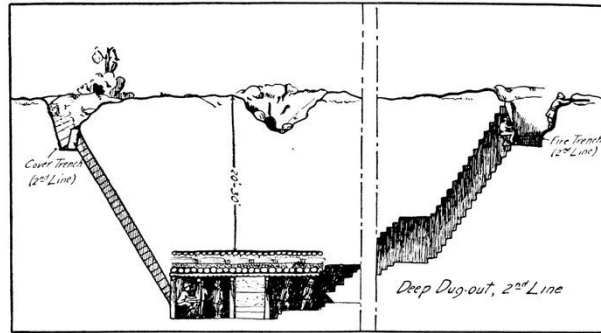
Liaison must be assured by other means than the telephone as this latter system usually breaks down under severe and continuous shelling. When the telephone is out of order, chains of runners must be resorted to. In the ordinary life of the trenches runners are not used. For this reason, when an attack is expected, new chains of runners must be established.



Supplies in the depots must be kept up to contemplated requirements. Besides the regular depots of the sector, numerous small ones are now established for the better distribution of extra ammunition and for the supply of reinforcing and counterattack troops. Supply parties are detailed to carry up ammunition during the battle.

Moral Preparation of the Troops. When an attack is expected, this fact must be told the garrison of each area, so that it can better prepare for it. The troops must not be left in ignorance, and upon the officers informing them devolves the responsibility of performing this task in such a manner as to raise the morale of their troops. The men must be maintained in the best of physical condition. Their food supply must be regular, of good quality, and sufficient. The tour of duty of the interior relief of the first line is shortened.

Duties during the Preliminary Bombardment. The most important and final act in preparation for the attack is a heavy and unusual bombardment of the position. Concentrated fire is brought to bear upon all visible points of the sector to demolish trenches, emplacements, dugouts, routes, and ammunition dumps, and to destroy the barbed wire entanglements. Certain batteries are detailed for counter-battery work. Barrages are established to prevent reinforcements from coming up and to cut communications with the rear. Against this bombardment, the defense has both a passive and an active rôle to play.



Maintenance of the Garrison during the Bombardment. The violence of preliminary bombardments is ever increasing and there seems to be no limit. High explosive shells of every caliber are used against the position. Any element of your system that is not properly concealed and can be located by the enemy is subjected to concentrated artillery fire until it is blown out of existence. Casualties inflicted upon the garrison are severe—in fact, it would seem from the violence of the artillery fire that few of the garrison could escape. However, when the assault develops, a machine gun may be found intact here and there in the position. One or two of these powerful weapons, by the use of flanking fire, may be enough to cause the failure of the enemy's attack. It is possible to say that upon the use of M. G.'s depends the whole interior defense of the sector.

The maintenance of the garrison during the bombardment depends, of course, upon the number and depth of dugouts. The service of defense, however, requires many of the troops to execute their duties in the open trenches. The leaders must set the example in this respect by making their rounds, and taking their posts to observe the area from the observation posts. In other words, they must not stay in their dugouts.

The bombardment naturally causes a great deal of repair work to be done. This is executed at night or during lulls in the shelling. Only emergency works, however, are executed, such as repairing of accessory defenses, cleaning of trenches and rebuilding of shelters. When this work becomes too heavy for the garrison, reserve troops are brought up to assist.

Reply with Artillery Fire. The only means to decrease the effectiveness of the hostile bombardment is to reply with your own artillery. For this purpose, a certain part of your artillery is detailed for counter-battery work. The emplacements or approximate locations of the hostile guns are set

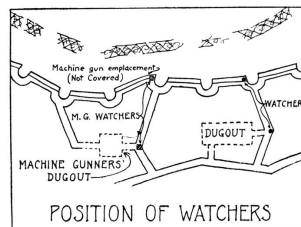
down on maps. These are divided amongst the counter-battery artillery who attempt to silence or neutralize the same.

In a sector, the commander may ask for what is known as counter-preparation fire. This fire is similar to the preliminary bombardment of the enemy. It covers the entire position of the enemy and is just as methodical in its scope. All the artillery that is at the disposal of the general is used for this purpose, and consequently the call for this fire must be through the general.

Similar to the counter-preparation fire, but on a smaller scale, is what is known as preventive fire which may be called for from the support artillery. This fire covers the first and second lines and is used to prevent the enemy from assembling his troops in the first line for the assault. This fire has been found very efficient. It is called for by telephone and not automatically by rocket.

II. Defense of the First Line during a Hostile Attack

Duties of Watchers. At the moment the assault is launched, part of the enemy's artillery establishes a barrage on your first line. The rest of the artillery, however, continues the regular bombardment to deceive the defenders as to the exact time of launching the attack. Therefore, the exact moment that the enemy comes over the top can only be determined by the watchers in your first line. It is their duty to give the alarm so that the garrison can reach its place in the firing line before the enemy reaches the same. This is a matter of seconds and not of minutes. Consequently, each dugout has a watcher located at its entrance and machine gun dugouts maintain special watchers. For each watcher at an entrance there is a second watcher within sight and calling distance ready to receive the alarm from the firing trench. If the second watcher is killed, the watcher at the entrance moves up and takes his place while another man from the dugout goes on watch there. These men are on duty only from fifteen to thirty minutes at a time. By a system of relief each occupant of a dugout serves a tour of duty as a watcher. The post of the watcher is protected as far as possible.



The Officer and N. C. O. of the Watch. During this time, the officer and the N. C. O. of the watch constantly make their rounds to see if the watchers are performing their duty efficiently. Instantly the alarm is given, the first duty of the officer of the watch is to set off the rockets calling for the barrage. At the same time, the orderly that accompanies him runs to the command post of the captain where the barrage call is confirmed by telephone and by rocket. Rocket signals are repeated and relayed from the same area until the barrage is obtained.

Defenders of the First Line. Immediately the call to arms is given, the troops of the first line spring out of their shelters and take their places at the parapet of the firing trench. If these parapets and fire trenches have been destroyed, the troops are not in immediate grave danger, because at this moment the enemy's barrage on your first line has lifted and is progressing toward the second line. Besides this, the assaulting troops can not subject the defenders to fire during the advance. Also, the assaulting column is subjected to the defensive barrage and is more or less in confusion, due to this fire and having to cross the shell-torn area of "No Man's Land." Then, too, what is left of your barbed wire entanglements will hold up the enemy's troops. In short, the advantage at this particular point lies entirely with the defenders if they can occupy the first line in time.

Among all the weapons of defense, the most powerful is the machine gun. The successful repulse of the hostile attack depends to the greatest extent upon the ability to use machine guns after the attack is under way. The opportune fire of one machine gun on the flank of an assaulting column may disorganize it and drive back its troops.

The Captain. As soon as the alarm is given or it is known that the enemy's attack has started, the captain throws his reinforcing platoons into the first line. These troops, in going forward, employ the boyaux laid down in the plan of defense. The hand grenades that they carry are stored in their dugouts. The chiefs of the reinforcing platoons do not necessarily wait for this order of the captain, but act upon their own initiative in sending forward their units. If, however, the first line has already been taken, the reinforcing platoons automatically carry out the functions of a counterattack by leaving their boyaux and going over the top to retake the captured trench. This last movement of over the top by the counterattacking troops is facilitated by the fact that the hostile artillery barrage has by this time passed to the rear of the first line.

The next duty of the captain is to report to his chief of battalion the alarm, which is done by means of a chain of runners, by signaling, or by use of carrier pigeons, because usually by this time the telephone system has broken down under the artillery bombardment.

III. Fighting in the Interior of the Position

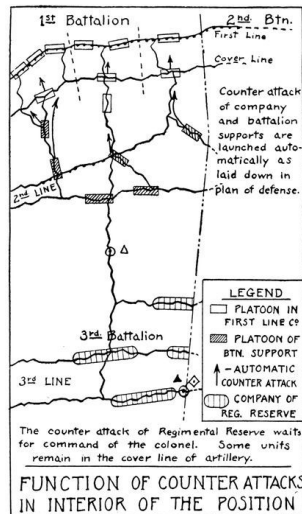
If the hostile assault on the first line is successful, the fight for the occupation of the position, which takes place in the interior, has only begun. Opposing his frontal progression through the position, the enemy will encounter, as we know, a series of defensive lines disposed in depth. Also, for instance, if a nest of resistance holds up the progress of a part of the assaulting troops, the successive overlapping waves will carry on along the flanks of this stronghold. If, then, the position is strongly divided into compartments, the fire of such defenses will take these overlapping troops in the flank and rout them. The defenders may also play an active rôle and the enemy is likely to encounter new troops sent up for the purpose of counterattack.

The division of the ground in support points, centers of resistance and sectors, has precisely for its object the localization of the enemy's attacks. Each area must be organized to defend itself, independently of any other part of the position. Consequently, the officers must explain this condition to their men, so that they will have no concern if they see that the enemy has penetrated and is attacking them on the flank or rear. On the other hand, if several of these compartments hold their ground, the enemy may find himself surrounded in a certain area and cut off from the rear by barrage fire. Thus abandoned in a part of the position that he has momentarily taken, a vigorous counterattack will drive him out.

Defense of an Area. The defense of each area is made similarly to that of the first line, as these areas embrace the successive lines of the position. The areas range in importance from front to rear, the support point embracing the first line, the center of resistance the first two lines, and the sector all three lines of the position. The alarm of an attack is relayed by the area commanders, the captain to the battalion commander, who in turn communicates it to the colonel of the sector. Not only the attacked portion of the sector is affected by this alarm, but the whole sector, so to speak, takes up arms. Let us see what happens as a result of this.

All defenders of the area who have a special duty, immediately occupy their posts and remain there awaiting developments of the attack. For

example, telephone men will remain at their instruments, runners will repair to their relay posts, observers occupy their observatories, and chiefs and headquarters officers remain at their command posts. And above all, the machine gun crews will man the guns disposed for interior fighting.



At the same time, the different troops held in reserve in the successive lines are used to hold their own line, to reinforce the line in front, or to make counterattacks. When the alarm is given, all supporting reserve troops take their places in their own lines. In each support point, the platoons held in reserve are sent forward as reinforcements to, or to make counterattacks against, the firing line. Similarly, in each center of resistance the companies held in reserve are employed as reinforcements or to recapture the first line. If, upon arriving at the first line, these troops receive no orders for particular duty, they replace the reserve platoons of the support point. Likewise, the sector reserve of the third line is sent forward to or, in case the enemy has taken it, against the second line. The static or passive organization of the defense lies in the series of lines of trenches disposed in depth, but the active rôle of the defender is carried out in these successive waves of defense moving forward in definite limits to meet the attack, not only to repulse it but to inflict as much loss as possible to the enemy. This transforms the defensive combat into more nearly a fight in the open ground, with the spirit of the offensive.

The Counterattack. It follows that the last and, very often, the most decisive means of defense of the sector is the counterattack. In each center

of resistance, a counterattack has for its purpose the retaking of the first line when this is captured by the enemy. It is made by the battalion reserves located in the reserve line.

The sooner the counterattack is made after its necessity, the more effective it will be. It should arrive at the first line almost, one might say, at the same time that the enemy does. At any rate, the counterattack should strike the enemy by surprise while he is still in a state of disorganization. Consequently, in the plan of defense of the battalion, the counterattack must be foreseen and all its details must be laid down. All these details must be known and practiced by the troops who are to execute the counterattack, so that when the alarm is given the counterattack will start automatically without orders and be carried to a finish like a good piece of team-work.

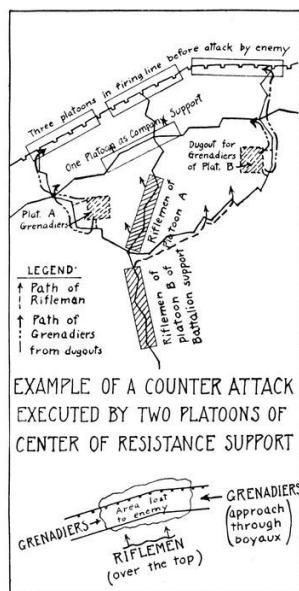
The counterattack is launched on the initiative of the leader of these troops, without waiting for the orders of the battalion commander. The latter may not know the exact situation nor be so located in the area as to appreciate the opportunity of the counterattack.

Troops Detailed for the Counterattack. The company, or companies, held as reserve of the center of resistance, are divided into half-companies or platoons under the command of a single leader. Each one of these detachments is assigned to a support point and is put at the disposal of the captain commanding the same, to counterattack on his first line if captured by the enemy. In order to secure the proper liaison between these two units, the counterattack troops send a messenger to the command post of the support point which they are assigned to. When the hostile attack is executed and counterattack troops are needed, the captain sends this messenger back to call them up.

Routes or Directions for Counterattacks. Each detachment of counterattack troops has a fixed route for its attack laid down in the plan of defense. Usually a boyau that it employs or along which it travels, fixes the direction. For this reason, in the interior of the position gaps are left in the barbed wire entanglements to permit the passage of these troops in the counterattack. Portable chevaux-de-frise are employed to fill up these gaps when necessary.

Form of the Counterattack. The counterattack may be executed as a frontal or a flank attack. Frontal counterattacks are, perhaps, not as effective as the latter, but they are more easily executed, especially in the case where

exact knowledge of the situation in the line in front is not known. This is the form of advance employed in the support point where the reinforcing platoons, starting forward to strengthen the line and finding it captured, change their tactics by leaping over the top and charging the line with grenades and bayonets.



A counterattack to the flank usually involves a little maneuvering. The most successful of these attacks are those executed in two detachments—for example, a party of grenadiers may attack the enemy occupying a part of the area either on one or both flanks. Its approach is usually through a trench or boyau. A second party of riflemen, at the same time, will make a frontal attack on the enemy over the top with bayonets. To completely surround the enemy, an artillery barrage is sometimes established to cut him off from the rear.

The battalion commander works out the plan of counterattack down to the most minute detail. But when the hostile attack is carried out, he is no longer the master of it. All the foreseen movements start at the proper time and are carried out in their workings like a piece of mechanism. The time of the start of the counterattack is in reality given by the enemy himself.

Repair of the Position. After a hostile attack has been repulsed, the chief of area must realize that similar attempts may be made without delay. The first and most important repair work that must be done is to construct a

parapet along the firing line. There may be no trench left along this line, but a fire parapet of sandbags must be constructed immediately. At night reserve troops are brought up with tools and sandbags to put the line in the best state of defense possible.

CHAPTER VI

ATTACK OF A POSITION

Phases of the Attack. As we have already learned, the attack of a position by a unit comprises three phases:

The preparation of the attack.

The assault against the first hostile line.

The exploitation of the success by fighting in the interior of the position for its occupation.

These phases are the same as those in open ground warfare, but their importance, aspect, and order are not the same. Especially the preparation of a trench attack is such a preponderant phase of the attack that upon the manner of its execution depends the success of the attack. The exploitation of the success is a long and decisive operation. In trench warfare, the assault is only the beginning of the fighting for a position, and its purpose is to make a breach in the enemy's first line. Through this breach, the reinforcing and reserve troops are pushed in behind the assaulting columns to deliver the real combat for the capture and occupation of the position.

Thus we see that the assault, which in open ground warfare is the final phase of the battle, is, on the contrary, in trench warfare the first move of the fight. In open warfare, the order of the battle is: the preparation, the approach march and infantry combat, and finally, the assault. An offensive in trench fighting consists of, first, the preparation; next, the assault against the first line; and last of all, the interior fighting for the position.

The cause of this reversion of the rôles of the attacking troops in a trench offensive operation lies in the strength and stability of the fortified front. It is impossible to maneuver against a continuous line which extends without a single breach. It is only possible to maneuver behind the first line or

within the position after a breach has been made. If a part of the first line gives way before the pressure of the assailants, the leader of the attacking units pushes forward his troops through this breach without consideration of the strategical value of the part of the line broken.

I. Preparation of the Attack

Front of Attack. The tactical unit for an attack is a division. A division that is placed in the front line for this purpose is called an attacking division. To such a division is assigned the task of attacking and capturing a definite length of front of the hostile position. All the necessary means to carry out this mission are in the hands of the division commander.

The tactical unit for an *assault*, in this division, is the battalion. Each assaulting battalion is placed in front of that part of the hostile line against which it is to operate, which is called its front of attack. The length of this front is variable with the tactical situation. Also, this length in no way indicates to the enemy the strength of the attacking troops because they are disposed for the attack in depth. According to the size of the front of attack, the battalion commander will place two or three companies in the assaulting column, and two or one company as reinforcing support.

Conditions of the Assault. An assaulting battalion must be disposed in a certain manner, which is called the assaulting disposition. In establishing this disposition, we must consider the distance of the assault and the outline of the assaulting lines.

Distance of the Assault. The distance over which the troops make their assault across "No Man's Land" must not be too great. This is to avoid as much as possible the enemy's barrage and machine gun fire. For this reason, parallels of departure or jumping-off trenches, if necessary, may be constructed out in front of the first line for the assembling of the assaulting troops. On the other hand, your own first line should not be so near to the enemy's position as to be in the zone of artillery dispersion when your own batteries are firing at the enemy's first line. However, one is not always master of this distance, as it is a result of long fighting and occupation of the two positions. The proper concealment of these attack works, such as the parallels of departure, is an essential condition of success.

The Trace of the Lines of Departure. The trace of the lines of departure must be parallel to the first line of the enemy, not to your own, so that the assaulting troops will simply have to start in a perpendicular direction from

the lines of departure to reach their objective. In other words, the assault is a simple frontal attack without maneuvers.

Ground Preparation for the Assault. It follows from the statements of the conditions of the assault and also of the tactical and material preparation of the attack, that the terrain from which these troops will start must be specially organized for this purpose. We have discussed the principles and details of the organization of the position for the defense, and it is clear that these must differ materially from those for the attack. An attack, like a defense, is made with the units disposed in depth. To launch an attack, certain of the defensive works that play but a passive rôle in the resistance, such as barbed wire entanglements, must be rearranged, modified, or partially eliminated. Other elements, such as routes, boyaux, and supply depots that facilitate the forward movement of reinforcements, are multiplied.

Details of the Preparation of the Ground for the Attack. All the details of the preparation of the ground for the attack are laid down in an order called the “Plan of Ground Disposition” issued by the general of the attacking division. This order is divided into two parts: the organization of the works, and their execution.

Organization. We know that the disposition of an assaulting unit is in the form of successive echelons, called waves. In order to protect these waves before the assault, it is necessary to construct for them trenches parallel to the enemy’s first line; and for this reason these are called parallels of departure. The trace of the first parallel fixes the directions of the others behind. For example, an assaulting battalion which is going to advance in four waves, might have two parallels of departure with two waves in each parallel. These parallels are narrow trenches like the defensive lines, but with a series of steps in the front side so that the men can go over the top easily. If there are no steps, trench ladders or footholds must be used.

Theoretically, one might think that the distance between these successive parallels should be the same as that between the attacking lines. But this is not practicable. Such practice would mark out boldly on the ground, for the enemy’s air observation, your intentions and your dispositions for the attack. Therefore, this method is strictly prohibited. Besides, it is not necessary, since the regulation distances between the lines and waves are

not realized from the point of departure of the assault. The object of the assaulting companies is to cross "No Man's Land" as quickly as possible to avoid the enemy's defensive barrage and machine gun fire. Consequently, when the different lines of the first wave jump out of the same parallel of departure, they will dash across to the enemy's first line with little regard to distances but with a fixed idea of reaching their first objective before the hostile garrison, and before the enemy's barrage intervenes. They attain their regulation distance as soon as the zone of barrage is passed and when they have arrived at and crossed the first hostile line.

In order not to print the attack on the ground, so to speak, these different parallels of departure are not constructed in front of your position unless absolutely necessary. They are constructed when the distance across "No Man's Land" is more than five hundred yards, when there is natural cover, such as woods to conceal them, and sometimes on reverse slopes. The different defensive lines of the sector make very convenient parallels of departure. The first waves of the assaulting company are placed in the firing and cover trenches of the first line. The reinforcing platoons are disposed in the transversal and intermediate and support lines. Behind these lines are constructed *places d'armes* or assembling places for the reserve troops. These *places d'armes* consist of a series of short transversal trenches leading off both sides of a central boyau.

If the first hostile line is too far from our own first line, on the night previous to the attack, hasty parallels of departure are constructed out in "No Man's Land" for the leading wave of the assault. This is to give this leading element the best possible chance to arrive at the enemy's first line uninterrupted by artillery fire and before the hostile garrison.

The work of preparing the ground for the disposition of the attacking troops also comprehends the following:

Several boyaux.

Command posts and observatories in advance of those of the defensive sector. Each chief must now be in the middle of his unit and not behind it.

Depots and medical aid stations. Each company has its own little ammunition depot near the post of the captain.

Ditches for telephone wires. Each battalion has telephone wires brought up in its main boyau as far as the first parallel. At this point, telephone

material is stored so that the system can be carried into the enemy's position with the least delay.

Small bridges are constructed for the passage of the parallels by the reserves.

It is not necessary that the shelter accommodations be numerous, especially in the first line, because the assaulting troops are brought in only a few days before the attack.

The Execution of the Works. The second part of the plan of ground disposition contains the details of the execution of the works for the preparation of the ground previous to the attack. This part consists of;

1. Emergency works (necessary to the assault).

Observatories and command posts.

The first two parallels of departure for each battalion. If none need be constructed, the defensive lines used for this purpose are designated.

Secondary boyaux between parallels, one for each company.

Entrance and evacuation boyaux.

Ammunition depots and water points.

Dressing stations.

If there is more time, the following are constructed:

The third and fourth parallels.

Telephone wires and material brought up.

Bridges for crossing the parallels.

2. Time of duration of the execution of the works.

This may vary from three to twelve days, or even more, depending upon circumstances.

3. Division of labor.

This depends upon the number of special working parties at your disposal. A whole division may be assigned for this work, but it is not the division that will make the attack.

4. Tools.

The tools required for this work are requisitioned from the headquarters of the attacking army.

5. Ammunition depots.

Cartridges, hand grenades, and signal rockets.

6. Ammunition depots.

Sandbags and logs.

7. Transportation of supplies.

Special means of transportation besides the usual ones, railroads, motor trucks, mules, horses, etc.

Artillery Preparation

During the time employed in preparing the ground for the attack, the artillery executes the preliminary bombardment. This comprehends three kinds of fire.

1. **Counter-Battery Fire.** A certain part of the artillery is detailed to destroy the hostile batteries or, if this is not possible, to neutralize them. This result is obtained by methodical fire on precise targets. Each battery or group of batteries is assigned certain emplacements to destroy. Counter-battery fire is long-winded work and is begun several days or even weeks before the attack. If the hostile batteries are not destroyed before the time of attack, they must be neutralized at this moment by violent shelling by all the batteries disposed for this purpose, with shrapnel and other special shells. This counter-battery action will hinder the hostile artillery from executing defensive fire, barrage, preventive fire, counter-preparation fire. It is carried out by special groups of heavy and light artillery under control of the general.

2. **Fire on Communication Routes, Depots, etc.** This fire, besides its destructive effect upon the enemy's position, hinders the arrival of reinforcing troops, material, ammunition, and food. The shelling of distant roads, depots, and bivouacs is carried out by batteries of long-range guns. The fire on the nearest communications, such as interior supply routes, entrances of boyaux, kitchen emplacements, etc., is executed by light batteries of the divisional artillery.

3. **Destructive Fire.** Before the attack, the whole of the enemy's position is submitted to methodical and violent artillery fire for the purpose of destroying:

Obstacles which may hinder the advance of the assaulting column: accessory defenses, such as barbed wire entanglements.

Elements of the defense, such as strong points, machine gun emplacements, observatories, depots, shelters, and dugouts.

Each different kind of target calls for a certain number of rounds of a particular caliber. These calculations of the different numbers of rounds are the results of experience and are set down in tables for the information of the sector commander. The time, therefore, necessary for a proper preparation for the attack is a function of the number and strength of targets. The artillery preparation may last from one to twenty days, but it must continue until the elements of the hostile position are sufficiently destroyed to assure a successful attack.

Before an offensive, there is placed at the disposal of the attacking divisions, besides their regular divisional artillery, a special allotment of batteries. The different calibers of guns are used according to the nature of the target. The largest calibers are employed against the strongest elements of the organization: nests of resistance, strong points, and deep dugouts. Light artillery is used to obliterate trenches and boyaux and interior barbed wire entanglements. Trench mortars serve to destroy the elements of the first line and its accessory defenses.

Plan of Artillery Action. The details for the execution of these different artillery fires are laid down in the plan of action of the artillery of the army corps or of the division. This plan is a part of the plan of battle of the division. These details must be carried out punctually and with precision. The sector commanders are not interested directly in the execution of these fires, but they are in their results.

Duties of the Infantry during the Artillery Preparation. In connection with the artillery fire, the infantry in the position must observe the results of the bombardment and the leaders report their opinion upon its control.

Observation. The observation of the counter-battery fire, long-range fire on communication routes, and destructive fire on the interior of the position, falls upon the artillery's ground observers and the aviation service. But the duty of observing the results of destructive fire on the enemy's first line is carried out with the help of the infantry observers. Each regiment must observe the front upon which it is to make the attack. Special attention is paid to the destruction of barbed wire entanglements and machine gun

emplacements. The interested infantry does not content itself with a passive observation, but must send out at night, or even in daylight, offensive reconnaissance parties. Their mission is to actually go into the enemy's first line, to ascertain its condition, and to investigate the breaches in the hostile entanglements. The tactics of such a reconnaissance are the same as for a raid. The information gathered is sent to the division headquarters in the daily report of the intelligence officer or in special information reports.

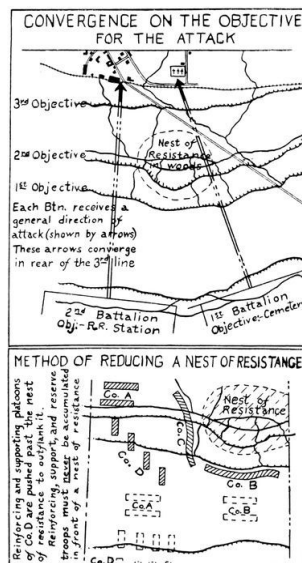
Control. It is both the privilege and the duty of the infantry commanders to state in their daily reports their opinion on the artillery preparation. If they do not express their exact opinion on the progress of the bombardment, they are liable to pay for their carelessness at the time of the assault.

Destruction of the Enemy's Position. The experiences of three years of trench warfare have given bitter proof that an attack against an insufficiently destroyed position will not only fail but will result in great losses. It is sheer folly to attempt an assault against undestroyed entanglements, or against a line in which the machine gun posts have not been demolished. One does not oppose materiel with men. In trench warfare, the conquest of the ground is made by the artillery; the infantry follow up this conquest to occupy and to hold the ground. The essential condition for the success of an attack depends upon the proper destruction of the elements of a position by the artillery.

Plan of Battle. The plan of battle depends upon, first, the strength of the enemy, obstacles to the attacking troops, lines of defense, nests of resistances, barbed wire, etc.; and secondly, the mission of the attacking unit and the means at its disposal.

Strength of the Enemy. The Information Service of a division has so many different methods of obtaining information and data of the enemy and his defenses that the assaulting troops can be constantly kept advised of the state of the hostile position. The regiment receives daily from the division headquarters an information bulletin, aëroplane pictures, and precise maps of the hostile lines. Upon these data the regimental commander bases his plan of battle. Comparisons of aëroplane pictures and maps are particularly useful. From all these data, the regimental intelligence officer makes pertinent extracts for the information of the battalion commander. Copies of the different maps are distributed by the battalion and company

commanders to their officers and non-commissioned officers in order that they may have a better knowledge of the hostile position for the attack.



Mission. The general plan of battle is issued by the division commander. Based upon the terms of this order, the Colonel, Battalion Commander, and Captain issue detailed orders of their own plan of battle. The plan of battle of a small unit usually comprehends the following points:

Mission of the unit in the attack.

Front to be attacked, with definite limits.

Objectives, different lines to be taken, the last objective.

Disposition of the unit for the attack; formation, intervals.

Disposition of the unit in the parallels of departure before the attack.

The day (D) and hour (H) of attack.

Direction of the attack; compass angle.

Position of the leader.

Tactical relation with the neighboring units during the advance.

Use of specialists; machine guns, trench cleaners, etc.

Support of artillery during the attack.

Unit liaison during the attack:

With neighboring units and leaders (runners, telephone, signaling, carrier pigeons).

With the support artillery (detachment of liaison, observatories, and rockets).

With air service (rockets, signaling, Bengal fires).

With balloon service (searchlights).

Equipment for the attack.

Supply of ammunition and water; advanced supply points.

Evacuation of the wounded; dressing stations, routes of evacuation, auto ambulance points. Also, routes of evacuation for prisoners and assembling points for same.

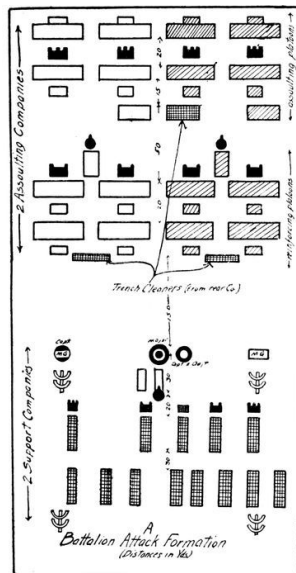
This plan of battle is so drawn up that it may be executed any later day that may be designated in the order for the attack.

Preparation of the Men for the Attack. The work of fitting their men physically, professionally, and morally for the task before them devolves upon the leaders of all ranks. By a system of relief, the men of the garrison get short periods of rest behind the sector to shake off the atmosphere of the trenches. Continuous instruction of both men and officers in the basic principles of fighting should be carried on at all times. The moral training of the men depends to a great extent upon the officers. The leaders, by example and encouraging words, should instill into their soldiers that fighting spirit which makes for success in battle.

II. The Assault

Formations for the Assault

Disposition in Depth. The assault has for its purpose the capture of the first hostile line, that is to say, the crossing of “No Man’s Land” and penetrating the enemy’s position. The assault is only the beginning of the combat. When the breach has been made in the enemy’s first line, the assaulting and reinforcing troops must continue the fighting in the interior for the conquest of the position. The direct object of the assault is to open the way for the attacking troops into the position. The attacking troops are disposed in depth in a series of echelons, so that during the advance each echelon is brought into the battle at the proper moment. This rule of formation in depth is followed without exception by every unit of whatever size.



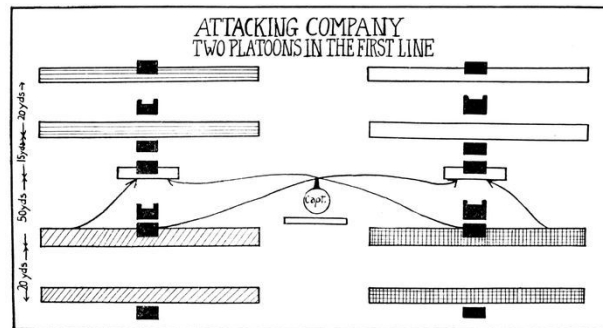
The regiment forms with one or two assaulting battalions, and two or one battalion behind in reserve. The battalion has three or, better, only two assaulting companies and one or two companies with machine guns, in the second echelon as support. The company may have three, but more usually

two, assaulting platoons with one or more reinforcing platoons in the second line.

Waves. The successive echelons have received the name “waves.” This designation is expressive but not clear, and causes mistakes.

First, waves may be formed by different dispositions: in deployed line or in line of small columns.

Secondly, the waves are not always composed of the same strength of troops or units, and consequently are not of the same disposition. The battalion commander will designate his first assaulting company as his first assaulting wave. Similarly, the company commander designates his assaulting platoons as the first wave; and the platoon leader, his first skirmish line as the first wave of the platoon.



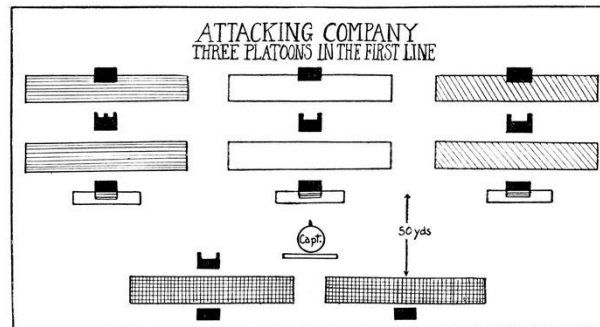
Lastly, the real significance of the term “wave” lies in its application to the formation of the departure of the attack and during the first part of its advance into the interior of the hostile position. As soon as the assaulting echelons are held up by nests of resistance, they are obliged to halt while neighboring echelons on both flanks continue the advance. Also, reinforcing units will come up from behind to outflank such a nest. As soon as such a condition arises within the enemy’s position, the disposition of the troops in waves is lost and the fight is continued in the best formation possible under the circumstances.

Formation of an Assaulting Battalion

Assaulting Companies. In an assaulting company, a certain number of platoons are placed in the first or assaulting line, and the rest in the second

line as reinforcing platoons.

The assaulting platoons deploy usually in two lines of skirmishers, the different specialists being assigned to places in these lines depending upon their functions in the attack. The line of assaulting platoons is known as the line of combat. A third line of grenadiers follows closely behind the assaulting platoon, and may be called a part of it. These grenadiers are furnished by the reinforcing platoons and are known as trench cleaners. Their function does not begin until the hostile line is reached, and it is to clear out the trenches of the first line of the enemy while the assaulting troops continue their advance in the open ground.



The reinforcing platoons are either deployed in lines of skirmishers or lines of small columns. These platoons constitute the line of reinforcement and follow the advance of the assaulting platoons at a distance of about forty to sixty yards. The machine guns detailed to the assaulting companies by the chief of battalion are placed with the reinforcing platoons and are usually carried on the flanks.

Considering the battalion as the assaulting unit and adopting the term “wave” to designate each platoon, we may say that the company of the first echelon of the assault is disposed in two waves. The position of the captain is in front of the second wave between the two reinforcing platoons. His liaison group remains with him.

Supporting Companies. The supporting companies of an assaulting battalion constitute the second echelon. These companies, also, are formed in two waves. These waves are formed like those of the reinforcing platoons of the assaulting company, either in deployed lines or in lines of small columns. Usually the battalion has two companies in the first line and two

in the second line. In other words, the battalion carries out the assault in two echelons of two waves each, the distance between the two echelons being from 200 to 300 yards.

In rear of the support companies and following their advance, are the remaining machine guns of the battalion and the 37 mm. platoon.

The position of the battalion commander with his liaison group is directly in front of the second echelon and between his support companies.

Disposition of Assaulting Battalion in the Departure Trenches. We have seen that for an attack the ground is specially organized for this purpose and the attacking troops are disposed in *successive parallels* or *places d'armes*. Usually this disposition is as follows:

The *two waves* of the assaulting companies are placed in the first and second parallels which under ordinary circumstances will be the fire trench and cover trench of the first defensive line. *Consequently*, the different lines of each wave will occupy the same parallel, but at the moment of the assault the successive departures of the lines will permit them to automatically gain the necessary distances to separate them in the advance across "No Man's Land." *For example*, let us assume that there are four lines (this is the maximum) placed in one parallel. The men are numbered from 1 to 4 in successive groups of four each. Each No. 1 man belongs to the first line; each No. 2, to the second line, and so on. The distance between similar numbers in the parallels is about four paces, giving the proper skirmish intervals to the line. At the given signal of the leader of the line, the No. 1's go over the top. The No. 2's follow at the prescribed distance ordered or at another signal given by their leader. The other lines follow similarly, so that the assaulting troops have the regulation interval in line and a prescribed (not always regulation) distance in depth.

The two waves of the support companies are placed in a third parallel or *place d'armes*. This third parallel is usually an intermediate trench, or special work, executed for this purpose, located between the first and second defensive lines.

Machine guns and 37 mm. guns are located on the ground in relation to their places in the advance during the attack.

The battalions which are kept as reserves according to the plan of battle of the sector commander, are disposed in *places d'armes* organized behind the parallels of departure. The colonel and his headquarters occupy a

command post directly behind the assaulting battalion from where he can witness the launching of the assault and the crossing of “No Man’s Land.”

Order of the Attack. The attacking troops take the disposition that has just been explained some days before the attack. This disposition is laid down in a paragraph of the Plan of Battle. Each leader of an attacking unit, with a map of the ground on which his area is specially outlined, must study his disposition on the ground previous to the arrival of his troops. Besides this, the leaders must pay special attention to the hostile position, verifying the direction of the attack, and identifying the successive objectives and the nests of resistance that will be encountered.

When the general who is in charge of the attack judges that the artillery preparation is sufficient, that the supply systems for the attack are properly organized, that all is ready and, in short, that the psychological moment has arrived, he gives his order for the attack.

Based upon the divisional order for the attack, the colonels, battalion commanders, and captains issue their own orders. The difference between the plan of battle and the order for the attack lies in the fact that the latter fixes all the details of execution of the general operations laid down in the former. But also, the order for the attack contemplates the first measures to be taken as soon as the assault is carried out, especially the extension of the liaisons (telephone lines) and communications (boyaux) from the first parallel of departure up to the enemy’s position.

The order for the attack contains precise details of the following points:

Exact hour of the attack; or signal rocket for the same.

Details of the successive objectives; different resistances that will probably be encountered.

Time-table of the moving barrage and the rate of its advance.

Prolongation of the telephone lines; particular wires to be extended.

Prolongation of communications; boyaux to be built at night across “No Man’s Land”; units detailed for this purpose.

Supply of ammunition and water in the captured position; points of distribution.

Departure of the Assault [The hour (H)]. We have now arrived at the most important and critical moment of the battle, the hour (H) of launching of the assault. A few minutes before the hour (H) the men take the places

assigned them in the parallels with strict orders to remain there. Bayonets are fixed on the rifles. Useless movements and noise of any kind are prohibited. Each man holds himself in readiness awaiting the signal of his immediate chief. Each chief of unit himself ascertains that his men are in readiness and in good condition for the assault. He sends all wounded back to shelter. The leader must impress his men with the confidence that he knows his duty, that he has foreseen and is prepared to meet all circumstances of the coming battle; The value of troops at this moment depends upon their leaders.

Support Artillery. During the time just preceding the hour (H) all the artillery which does not execute counter-battery fire is employed to protect the assault. A part of this artillery is given to each of the assaulting battalions as support artillery. Liaison and observation detachments are sent by this artillery to the infantry battalions to which they are attached. These detachments have the same composition and rôle as explained in the defense of a position.

The artillery support holds under its fire that part of the hostile line assigned to its infantry battalion. A violent fire is especially directed against the enemy's first line, the objective of the assault. Protected by this fire, the assaulting battalion moves forward. For a few moments after the launching of the first waves over the top, this fire remains on the enemy's first line. The protective barrage is then lifted and is established on the hostile second line, where it also remains for a certain time, according to the barrage timetable, and so on.

Departure of the First Echelon (Assaulting Companies). At the precise moment or signal the first wave of an assaulting company jumps out of its parallel of departure, the different lines of which it is composed being formed as previously explained. In a similar manner, as soon as the first wave is launched, the second wave goes over the top of its parallel. All the different lines of the different assaulting companies advance straight to the front towards their first objective. The rate of advance, although not a run, is as rapid as the circumstances of the torn up ground of "No Man's Land" will permit, at the same time maintenance of the formations being paramount. Success depends upon the proper location of the specialists in the formations of the waves, and this order must not be lost in the confusion that results from too much speed.

The result is that all the lines of the assaulting companies start almost at the same time from the two parallels of departure, and with shorter distances between the lines than that called for by regulation. This formation is not a drawback, but, on the contrary, works to better advantage. The first and most important thing to avoid at this time is the defensive barrage of the watchful enemy. This is established as soon as he is aware of the assault. If the reinforcing line follows closely behind the line of combat, it has a better chance of passing the danger zone before the enemy's barrage is established. A company caught under the enemy's barrage will lose, on an average, fifty per cent. of its effectives. Also, the assaulting companies must arrive in the first hostile line as quickly as possible behind their own barrage. The assaulting platoons, who set the pace, should arrive there almost "as soon as the barrage lifts." Troops that arrive at this opportune moment will surprise the enemy in his shelters or in the act of taking his place on the firing line. It is better to risk losing a few lives by your own barrage than to give the enemy time to man his first line.

Finally, the regulation distances between the different lines and waves will be attained during the progress after passing the enemy's first line.

Departure of the Second Echelon (Supporting Companies). When the first echelon, or assaulting companies, have crossed "No Man's Land," the battalion commander usually launches the second echelon, or supporting companies. By this time, the battalion commander will have an impression of the assault, and the distance which separates the two echelons is at this moment the one usually required (200 to 300 yards). In order to avoid the enemy's barrage and to profit by any lull in his fire, the battalion commander chooses the exact moment for the departure of the second echelon.

Advance of the Reserves. As soon as all the waves of the assaulting battalion have penetrated the enemy's position, the reserve battalions of the regiment leave their *places d'armes* and move forward through boyaux to the first parallels of departure. In this position they await the order of the colonel to advance into the enemy's position to join the combat for its conquest. The colonel awaits the first report from his assaulting battalion before giving the order for the advance of the reserves.

III. Fighting in the Interior of the

Position and Exploitation

Principle of the Fighting. The fundamental principle of the fighting in the interior of the position is that each attacking unit from the division down to the battalion receives a definite objective. This objective is usually an area containing the lines of the hostile position with precise flanking limits. The farthest line in the objective area must be gained by the units at all costs. Beyond this last objective, strong reconnaissance parties and patrols are sent to keep contact with the enemy.

Usually the attacking regiments, which carry out the struggle in the interior of the position, are assigned the mission to capture the third or covering line of the artillery. The continuation of the struggle beyond this line and the capturing of the enemy's guns is what is called "the tactical exploitation of the success." The troops that carry out this operation are the reserves of the division held out for this particular purpose. The plan of battle does not include orders for this action.

Details of the Interior Fighting. Let us follow an attacking regiment in its progress in the interior of a hostile position, and consider the use of the different echelons of which it is composed: assaulting companies, reinforcing companies, and reserves.

Assaulting Companies. The advance of the assaulting platoons regulates the advance of all the successive elements. These platoons, after crossing the first hostile line, continue their movement without hesitation. The rate of advance is, of course, variable with the difficulties of the ground and the strength of the hostile organization. For this reason it cannot be very rapid. The pace of the leading elements of the assaulting column is foreseen in the plan of battle (depends upon the strength of the enemy's position), and this becomes the rate of advance of the moving barrage. If no unforeseen and serious resistance is encountered, the assaulting platoons will be able to keep up with their protective barrage. Sometimes, however, they may halt for a moment in their progress under cover of some shelter to regain their alignment.

As a part of the assaulting platoon comes the trench-cleaner detachment. The trench cleaners are armed with hand grenades and trench knives. Their function is to enter the hostile trenches taken and, profiting by the surprise effect of the attack, to vanquish the last resistance and to make prisoners of the defenders still in the dugouts. They pay special attention to picketing all entrances, exits, and shafts of dugouts so that detachments of the enemy cannot come out and fire into the rear of the waves that have already crossed. If the different lines of the hostile platoon are very far apart, the number of trench cleaners is increased, a detachment dropping off in each line.

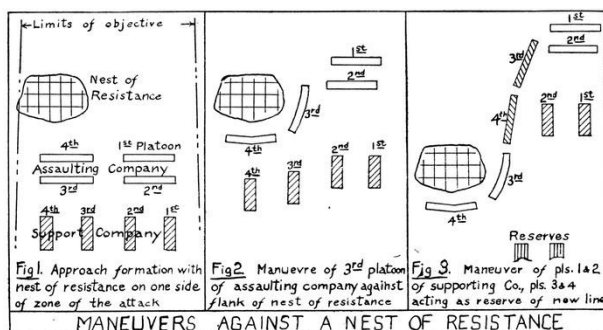
As long as the assaulting platoons do not meet resistance, the reinforcing platoons follow the former at the regulation distance to avoid mixing of the different waves of the attack. If, however, the assaulting platoons suffer severe losses, and are unable to continue the progress, the reinforcing platoon, either by order of the captain or upon the initiative of its leader, moves up and reinforces the combat line. Its function then becomes the same as that of the assaulting platoon.

Again, if a gap should occur in the combat line, due to losses or extension of the front, the reinforcing platoon moves in that direction and fills the gap. In general, the tactics of the reinforcing platoon are always to assure continuity of progress by outflanking resistances that have stopped the advance of the platoon in front.

The machine guns attached to an assaulting unit follow and protect the flanks. When halted they fire on retreating hostile troops and on special points of resistance. Their special use, however, is against counterattacks.

Reinforcing Companies. The reinforcing companies, with the machine gun reserve and 37 mm. gun of the assaulting battalion, after they have penetrated the hostile position, follow the progress of the assaulting waves at the prescribed distance. They advance in line of small columns but make use of any cover that the ground affords to minimize losses from hostile fire. During this advance, their mission is to rally all the elements of the assaulting companies and push them forward; and also to watch out for the flanks, pushing out to protect them on their own initiative if necessary. But when the line of combat is held up by resistance the supporting companies are maneuvered by order of the battalion commander. Naturally, the whole line of combat will not progress uniformly. Certain assaulting platoons will

be held up by nests of resistance, while others in the intervals will carry on with less interruption. Reinforcing platoons and support companies are maneuvered to outflank such nests of resistance and to fill the resulting gaps in the combat line.



Let us take a concrete example of an assaulting company followed by a supporting company held up by a nest of resistance (a ravine, reverse slope, or woods strongly organized with machine guns) and follow the movements of the reinforcing platoons and supporting company. (Examine the diagram showing maneuvers against a nest of resistance in connection with the following explanation.) The first figure shows the formation of the four platoons of an assaulting company advancing in the interior of a position followed by a supporting company. The 1st and 2nd are the assaulting platoons, and the 3rd and 4th the reinforcing platoons. The supporting company follows in double column of platoons. The second figure shows the 1st assaulting platoon held up by a nest of resistance with the 3rd or reinforcing platoon moved up on the right flank. Platoons 2 and 4 of the same company carry on, and this movement results in a gap in the line of combat between the 2nd and 3rd platoons. The third figure shows the 1st and 2nd platoons of the supporting company moved up in this gap, and the 3rd and 4th platoons acting as reserves. The nest of resistance is thus attacked from the front and flank and the continuity of the combat line maintained. When a nest of resistance is encountered the attacking troops must not accumulate in front, as this will multiply the losses. The line of combat will attack the front of a nest of resistance while the successive reinforcing and supporting waves maneuver against its flanks. The battalion commander will use all the weapons at his disposal as machine guns, 37 mm. guns, and rifle grenades to reduce the nest of resistance. If this method

fails he will have to call on the artillery for help, but this necessitates a modification of the artillery barrage time-table.

Rôle of the Artillery during the Attack. During the attack the artillery carries out the following different fires:

Counter-battery fire.

Protective fire on the flanks of the attack.

Accompaniment or barrage fire.

The first two fires are executed by a part of the artillery directly under the chief of artillery of the division, who in turn is under the control of the general.

The third kind of fire is executed by the support artillery, detailed to the different assaulting battalions as in the defensive organization. The action of this artillery directly interests each infantry chief of area, to whom are sent liaison detachments and observers (an officer, N. C. O.'s, and artillery orderlies). This third kind of fire is the only one that we will consider in detail.

Principle of the Accompaniment Fire. During the progress of the attack the support artillery establishes a moving barrage in front of the advancing infantry. This barrage travels by bounds. It remains for a certain time on each defensive line and at intervals between them so as to thoroughly sweep the intervening ground. The time-table, or rate of this moving barrage, is laid down in the plan of action of the artillery. The infantry leaders of course are acquainted with this time-table.

Other Means of Controlling the Barrage. The moving barrage may also be advanced successively by order of the commanding general according to information received from the leading elements of the assaulting battalion, from the artillery or aeroplane observers.

Calls may be made directly from the line of combat by rockets, or other means of liaison. With reference to the barrage there are two uses of rockets; one is for calling the barrage in the defensive, and the other is calling for an advance of the barrage during the attack. It may be remarked that it is impossible to call for a decrease in range of the barrage as this, of course, would be dangerous to your own troops. It is better to decrease the rate of advance of the barrage rather than to have it move too quickly and risk abandoning the infantry.

Lastly, the preceding means for controlling the barrage can be employed in conjunction. That is to say, a time-table for the barrage is adhered to during the first part of the advance when the positions of the hostile lines are well known, while during the last part, when the infantry's progress is not so regular and the ground is not so well known, each advance of the barrage may be called for by rocket, as each resistance met with is disposed of. If an assaulting battalion meets a resistance which it cannot reduce by its own means, it must call upon the support artillery for help. To do this, the artillery must halt its barrage and for an interval the time-table is not in effect. Such an emergency calls for a special action on the part of the support artillery, and also for the intervention of any other artillery at the disposal of the general. This particular action of the artillery may be called for automatically by rocket or by the other means of liaison. The rôle of the officer and N. C. O. of the artillery liaison detachment is to give the artillery the precise technical information necessary in such a case to establish its fire on a nest of resistance. This action of halting the barrage and giving the artillery a new objective is exceptional and delicate to execute. As soon as the resistance is reduced, the time-table is resumed.

Liaisons During the Attack. The proper co-ordination of all the foreseen developments of the attack lies in a good system of liaison. There must be leadership liaison for the transmission of orders and reports, and infantry-artillery liaison for the proper co-operation of the two arms. The means of liaison in the offensive are the same as those considered in the defensive organization. The principle of their use is the same but made more difficult by the progress of the different attacking echelons. It is the duty of all leaders to foresee the establishment of liaison for the engagement in their plan of battle, and also for the prolongation of these means with the advance of the attacking troops.

The means of liaison are:

Telephone lines (for leadership in artillery. They are lengthened from the parallel of departure to the interior of the enemy's position.)

Signal communication (established from the enemy's position to a central point in the rear).

Rockets (precise signification of each kind laid down in orders).

Carried pigeons (headquarters that are to carry these laid down in orders).

Detachment of special runners (essential to be established between all units).

Aviation. During the advance, a particular system of liaison is established with the air service. Each division has at its disposal a certain number of accompanying planes that fly low and follow the assaulting battalions. Their duty is to establish communication between the leading elements of the advancing infantry, and the artillery, and the commanding general. The infantry signal to these airplanes by means of rockets and Bengal fires, and the planes in turn communicate with the artillery and the commanding general by means of wireless.

CHAPTER VII

TRENCH ORDERS

The following outline gives a synopsis of information required in various paragraphs for Battalion Orders in Trench Warfare, followed by models of each.

Six orders are given in these models, viz.:

- I. Plan of Attack.
- II. Order for the Attack.
- III. Preliminary Order for a Relief.
- IV. March Order for Relief.
- V. Plan of Advance.
- VI. A Raid.

I. Plan of Attack

(Issued several days before attack)

1. **Rôle of Battalion**, *i. e.*, general description of offensive and whether battalion is in first line or reserve.

2. **Zone of Attack**. Accurate description of boundaries of hostile position to be attacked.

3. **Objectives**. Number of trenches to be assaulted and trench at which offensive will halt.

4. **Disposition for the Attack**. Companies in assaulting line—companies in support—apportionment of objectives to companies—distance between assaulting and supporting companies—disposition of M. G. Co. and 37 mm. gun—location of battalion C. O. during assault.

5. **Disposition of Units in Parallels of Departure**. Describe accurately trenches each unit will start from: *e. g.* Assaulting Cos.—first line and cover trenches. Supporting Cos.—intermediate trenches or *places d'armes* nearby starting points of battalion C. O., M. G. Co. and 37 mm. gun.

6. **Direction of the Attack**. Landmark to guide on, also compass direction.

7. **Tactical Relation with Neighboring Units**. Troops on flanks of battalion and responsibility for liaison therewith. Patrolling to establish connection if necessary.

8. **Artillery Support during Attack**. Barrage to start at hour H. Where and how long it will halt beyond the last objective.

9. **Liaison during Attack**. Usual liaison with artillery and aëroplanes—telephone lines to be established during attack—signals such as Bengal lights to show progress of assaulting companies.

10. **Supply**. Detachments (usually from supporting companies) to carry ammunition. Hour at which and points from which they will start.

11. **Evacuation**. Evacuation point for wounded and assembly point for prisoners.

Details **not** stated in this order. Time of attack—detailed instructions for barrage. Liaison to be established after attack—equipment to be taken. Information of the enemy not given as all officers have maps of enemy's trenches as well as of their own trench system.

II. Order for the Attack

(Issued night before attack is to be made)

1. Time; *i. e.*, hour H.
2. Further details of assault than given in previous order.
3. Further details concerning barrage, *i. e.*, Halts to be made by barrage—rate of travel.
4. Liaison to be established **after** attack.
5. Supplies to be carried.

III. Preliminary Order for a Relief

(Issued night before reconnaissance; *i. e.* 24 to 36 hours before relief)

1. (Based on regimental order.) Battalion to be relieved—Sector.

General Description of Sector: Centers of resistance—troops in each—troops on right and left of sector—reserve-command post of Colonel, Command Post of artillery support Commander.

Note: Battalion needs know no more about artillery support than location of command post. Details as to location of guns are not given out.

2. **Disposition of Battalion in Center.** Accurate description of: First line and units occupying it—intermediate line and units occupying it—support line and units occupying it—Battalion Command post—stations of machine gun company—dressing station.

3. Hour at which usual reconnaissance will be completed—each officer to reconnoiter area assigned him in 2.

4. Hour at which Major will assume command of center of resistance.

IV. Order for the March of the Relief

(Issued after reconnaissance and several hours before relief starts)

1. Dispersion point fixed by Colonel—Battalion's place in regiment—hour at which battalion clears dispersion point.
2. Relief to be made as per preceding order. Order of march of units in battalion. Routes to be followed and formation taken from dispersion point to point of rendezvous of guides.
3. Hour at which the guides will be at rendezvous. Location of rendezvous. Routes by which guides will conduct various units to their places.
4. Hour and place ration details will report to N. C. O. from battalion headquarters to go to kitchens for rations. Hour at which daily requests for material and ammunition will reach Major. Whether Major has depot to fill emergency requests.
5. Reports to be submitted by Captains. Usually written report upon occupation of area and daily reports submitted at same time as requests for material.

V. Plan of Defense

1. Direction and nature of probable attacks.
2. Plan of Defense of center. Duties of first line Companies. Orders for machine guns. Platoons to support machine guns.
3. Counterattack. Platoons for counterattack—their routes—reinforcing platoons to replace them.
4. Artillery Support. To establish barrage in “No Man’s Land” at usual (rocket) call.
5. Supply. Extra grenades or ammunition to be carried by counterattack or reinforcing platoons.

VI. A Raid

1. Date, hour, objective, and purpose of raid.
2. Composition of and company furnishing raiding party.
3. **Assistance to Raiders—Details of Execution of Raid:** *e. g.*, Grenadiers to protect flanks. Objective. Direction of attack. Preparation of explosives to destroy enemy's wire. Rocket signal for blowing up wire, rush of raiders and start of artillery or rifle grenade barrage. Duration of raid. Direction and return of raiders.
4. Duration and location of barrage.
5. Supplies. Raid is generally made by men from an organization in rear (support or reserve). Captain of first line company provides the material: Grenades, tools, explosives, rockets, etc.

I. Plan of Attack

1st Bn. 3rd Inf. 47th Div.,
Bn. Command Post,
1 Nov. 17, 2 p. m.

Field Orders

No. 7

Map

BERRY-AU-BAC, N. O.

1. **Rôle.** In the attack this Bn. will be in the first line. It will attack the west slope of the ENCLUME WOOD in a general offensive in conjunction with the Bns. on both flanks.

2. **Zone of Attack.** The zone of attack will be limited as follows: On the left by the road from TEMPLE FARM to CORBENY (exclusive), on the right by the imaginary line marked by the points, boyau ST. POL, point 6909 (first German line), point 7109 (cover trench, first line) hill 69.2 point

7216 (German support line) and topographical point 78.8, all these points inclusive.

3. Objectives. 1st. First German line, Plaine trench. 2nd. Cover trench, ENVER PASHA trench. 3rd. Support trenches, L'ENCLUME trench on the east and MARTEAU trench on the west.

On this last objective the assaulting companies will halt, maintaining contact with the enemy by means of contact patrols, especially on the right front towards the FORGERON WOOD and the west edge of the L'ENCLUME WOOD.

4. Disposition for the Attack. The Bn. will attack with two Cos. in the assaulting line and two Cos. in support. Assaulting Cos. (A and B) will take the usual assaulting disposition. Each Co. will be reinforced by a M. G. platoon.

The zones of action of the assaulting Cos. will be divided by the line: Point 6502 (ANSPACH trench), point 6606 (in the BONNET PERSAN WOOD), and the boyau PARSEVAL from point 6712 to point 6919. All these points to be included by the left Co.

The distance between assaulting Cos. (A & B), and supporting Cos. (C & D) will be 300 yards.

The remaining sections of M. G. Co. and the 37 mm. gun will follow the left supporting Co. (C) at 100 yards.

The Bn. C. O. will be located in front of the reinforcing companies to the left of the center, on the general axis of march, the boyau PARSEVAL.

5. Disposition of Units in Parallel of Departure. The assaulting Cos. will use for their parallels of departure the fire and cover line trenches between the boyau ST. POL and LIEVIN (inc.).

The supporting Cos. will use for their parallels of departure the ANSPACH trench, and the *places d'armes* built in rear of this trench and on each side of the boyau ST. OMER and ST. POL.

The Bn. C. O. with his headquarters, will start from the point of intersection of the ANSPACH trench and the ST. OMER boyau.

The M. G. reserve and 37 mm. gun are located in the 520 M. trench.

6. Direction of the Attack. The general direction of the attack is on the R. R. station at south edge of CORBENY, compass direction N. 10° E.

7. **Tactical Relation with Neighboring Units.** Left: connecting with 2nd Bn. 3rd Inf. on the road. Permanent liaison will be assured by the assaulting Cos.

Right: connection with 2nd Inf. at the points 6909 and 7216.

The right supporting Co. will take a formation in echelon with the right element refused, this latter element establishing connection with the troops to the right during the advance.

Reconnoitering patrols will also establish connection with the troops on the right in the L'ENCLUME WOOD.

8. **Artillery Support during Attack.** The assaulting waves will be protected by a moving barrage which will start at the hour (H). The moving barrage will halt and remain on the line 200 yards beyond the last objective at the hour (H plus 50 minutes).

9. **Liaison during Attack.** Usual liaison with the artillery and airplanes (rockets and flags).

As soon as the assaulting Cos. arrive in or halt before the third objective, a telephone liaison will be established between the Co. commanders and the Bn. commander.

A Bengal line of fire will be lighted by the assaulting Cos. when they reach the 2nd and the 3rd objectives.

10. **Supply.** Each supporting Co. will send a party often men to the TEMPLE FARM as supply detachments. These detachments will start with ammunition supply at the hour (H plus 1 hour) for the command posts of the assaulting Cos. in the captured position.

11. **Evacuation.** Evacuation point for wounded will be at RIVOLI CENTER (northeast edge of DE BEAU MARAIS WOOD). Assembly point for prisoners will be at TEMPLE FARM.

Major, 3rd Inf.,
Bn. Comdr.

II. Order for the Attack

1st. Bn. 3rd Inf. 47th Div.,
Bn. Command Post,
4 Nov. 17. 7 p. m.

Field Orders

No. 8

Map.

BERRY AU BAC N. O.

1. In accordance with Field Orders #7, these headquarters, the attack will be made at the hour (H).

H = 10:30 A. M.

2. The three objectives of the attack are: 1st, PLAINE trench; 2nd, ENVER PASHA trench; 3rd, L'ENCLUME and MARTEAU trenches.

For the capture of these three objectives the plan of the Bn. commander is as follows:

The assaulting Cos., after taking the 1st and 2nd objectives will proceed to the attack of the 3rd objective, while the mopping up parties clean up the captured trenches. As the assaulting Cos. leave the 2nd objective, the Bn. commander will give the order for the supporting Cos. to start, in order that they will reach the 2nd objective at the same time the assaulting Cos. reach the 3rd. In case the assaulting Cos. are compelled to stop before reaching the 3rd objective, the Bn. commander will use the supporting Cos. to outflank the nest of resistance expected to be encountered in L'ENCLUME wood.

3. The moving barrage will halt 10 minutes immediately in rear of the 2nd objective and 10 minutes on the 3rd objective. With the exception of these halts the barrage will move at a uniform rate of 20 yards per minute.

4. As soon as the Bn. commander reaches the point 6919 a telephone liaison will be established from the starting point in the ANSPACH trench to point 6919.

From point 6919 a signal liaison will be established with the artillery observation post in the EDMOND BUTTE.

5. The supply detachments will carry 400 hand grenades, 50 illuminating and 50 signal rockets.

Major 3rd Inf.,
Bn. Comdr.

III. Preliminary Order for a Relief

Order for the Occupation of a Center of Resistance

Field Orders
No. 5.

3rd Bn. 80th Inf.
1 Nov. 17. 6 p. m.

1. In accordance with orders of the regimental commander, the 80th Inf. will relieve the 132nd Inf. in the TEMPLE sector during the night 2–3 Nov. 17.

In this sector two Bns. are in the first line in the centers of resistance BONNET PERSAN WOOD and PINS BUTTE.

The 3rd Bn. 80th Inf. will occupy the BONNET PERSAN WOOD center, which is limited on the east by the ST. POL boyau (inclusive) and on the west by the TEMPLE FARM-CORBENY road (exclusive) which road is the dividing line between the two centers.

Adjoining units:

On the right the 45th Inf., occupying EPINAL sector.

On the left the 1st Bn., 80th Inf., occupying PINS BUTTE center.

In reserve, 2nd Bn., 80th Inf., in the BEAU MARAIS WOOD.

Command post of the regimental commander (sector commander), TEMPLE FARM.

Command post of the artillery support commander (72nd A. C.), EDMOND BUTTE.

2. Disposition of the Bn. in the BONNET PERSAN WOOD center.

In the first line: Co. A in the firing and doubling trench (Baden trench) between the ST. POL and BONNET PERSAN boyaux (both inclusive), Co. C in the same firing and doubling trench between the BONNET PERSAN boyau (exclusive) and the LIEVIN boyau (inclusive).

In the intermediate line (ANSPACH trench): Co. B and the detachment of pioneers.

In the support line (520 M trench): Co. D and Bn. Hq.

Command post of the Bn. Cmdr. (Commander of the center), Command post of ST. OMER (in the ST. OMER boyau).

Three sections of the M. G. Co. will relieve same numbered sections of the relieved Co. in the different lines of the center. The remainder of the M. G. Co. with the 37 mm. gun will be kept as Bn. reserve in the 520 M trench.

Dressing station: in the 520 M trench at the intersection with ST. OMER boyau.

3. The reconnaissance for the relief will be made early in the morning of 2 Nov. 17 (usual composition) and will be completed at 10 a. m. Each officer will reconnoiter the area assigned to him in this order.

4. After the relief the Bn. Cmdr. will assume command of the center at 6 a. m. 3 Nov. 17.

Major 80th Inf.,
Bn. Cmdr.

IV. Order for the March of the Relief

Field Orders
No. 6.

3rd Bn., 80th Inf.,
2 Nov. 17.—12 o'clock noon.

1. In accordance with orders of Regimental Commander, the dispersion point of the regiment for entering the TEMPLE FARM sector will be

PONTAVERS. This Bn. will be the first Bn. to leave. It will clear PONTAVERS at 5 p. m.

2. The occupation of the BONNET PERSAN WOOD center will be made in accordance with Field Orders No. 5, these headquarters. The order of march of the Bn. will be: Co. A, Co. C, Bn. Hq., C. B., Dtch. Pioneers, Co. D, Mach. Gun Co., and 37 mm. gun.

The march from PONTAVERS to HALTE will be via the CORBENY road in column of squads with 200 yards distance between cos. From HALTE to point of rendezvous of the guides the Bn. will march along the right of R. R. track in column of twos with 200 yards distance between cos.

3. The usual guides will be sent from the relieved Bn. to be at the point of rendezvous by 6 o'clock p. m. The point of rendezvous of the guides will be south of EDMOND BUTTE at the intersection of the CORBENY road and the R. R. track.

From this point the cos. will be conducted by the guides to their respective positions in the trenches. Cos. A, B, and Pioneer Dtch. will use for their route the ST. POL boyau. Cos. C, D, Bn. Hq., M. G. Co. and 37 mm. gun will use for their route the ST. OMER boyau.

4. At 7 o'clock p. m. each night on and after 3 Nov. 17, the ration details from each organization will report at the point of rendezvous of the guides to a N. C. O. from Bn. Hq. who will conduct them to the kitchens at PONTAVERS for rations. Requests for material and ammunition will be sent to the Bn. Cmdr. every morning to reach him before 6:30 o'clock a. m. Emergency requests made at other times will be filled as far as possible from the Bn. Cmdr's depot.

5. Company commanders will make a written report to Bn. Cmdr. as soon as they have occupied their areas, and thereafter will make daily reports to be submitted with their requests of material.

Major 80th Inf.,
Bn. Cmdr.

V. Plan of Defense

1st Bn., 20th Inf., 43rd Div.,
Bn. Command Post,
29 Nov. 17, 2 o'clock p. m.

Field Orders

No. 10

Map.

BERRY-AU-BAC N. O.

1. **Direction of Probable Attack.** The enemy may possibly make a frontal attack starting from PLAINE and ENVER PASHA trenches and supporting it on the left from L'ENCLUME WOOD, or a flank attack starting from PLAINE redoubt for the purpose of striking our first line and ANSPACH trench from the rear by first penetrating the sector on our right.

2. **Defense of the Center of Resistance.** When the assault develops Cos. A. and C. in the first line, will maintain the defense of their own line by their own means. The two machine guns located at 6501 ANSPACH trench will be taken immediately to the right flank of Co. A, near POSTE DE NEMOURS and take up a position to fire to the north toward L'ENCLUME WOOD and to the east toward PLAINE redoubt. The C. O., Co. B, will place a platoon permanently in the ST. POL boyau at the junction of the ANSPACH trench, and, when the alarm is given will send it forward to occupy and hold the POSTE DE NEMOURS salient in support of the M. G. Platoon. When this platoon has moved forward, its place will be taken by 1 Platoon of Co. B, (located in ANSPACH trench between the ST. POL and BONNET PERSAN boyaux) which will protect the right flank.

3. **Counterattack.** The two remaining Platoons of Co. B are assigned to the commanders of Cos. A and C respectively for use in counterattack. One Platoon will employ the BONNET PERSAN boyau, the other the LIEVIN boyau. The Bn. Cmdg. will replace these two Platoons by two Platoons of Co. D, in the 520 M. trench.

4. **Artillery Support.** The supporting artillery will establish a barrage in No Man's Land in front of the first line. The barrage will be called by the usual signal.

5. **Supply.** The counterattack and reinforcing Platoons will carry 5 grenades per man in addition to their regular equipment.

Major 20th Inf.,
Bn. Cmdg.

VI. Order for a Raid

Field Orders
No. 12

1st Bn., 80th Inf.,
Bn. Command Post,
10 Nov., '17, 2 p. m.

1. At 3 o'clock a. m., 20 Nov., '17, a raid will be carried out against the western salient of PLAINE TRENCH for the purpose of capturing prisoners.

2. The raiding party will consist of a lieutenant, 10 hand grenadiers, and 20 riflemen from Co. D, 80th Inf.

3. Two parties of grenadiers each will be furnished from the same Co. to protect the flanks of the raiding party on each side of the salient. The raiding party will penetrate the enemy's first line by a frontal attack, capture the defenders of the salient and return directly by the same route. The raiding party will crawl into "No Man's Land" in front of the PLAINE salient until it reaches the enemy's barbed wire and will prepare detonators for the necessary breaches. The signal to explode the detonators and rush into the salient will be given by the leader of the raid by rocket. The raid will last ten minutes.

4. For ten minutes after the rocket signal the artillery will maintain a barrage on the flanks and in the rear of the PLAINE salient to protect the raiding party.

5. Co. C, which occupies the first line in front of the PLAINE salient, will provide a supply of 60 detonators, 200 hand grenades, and 3 signal rockets for use of the raiding party.

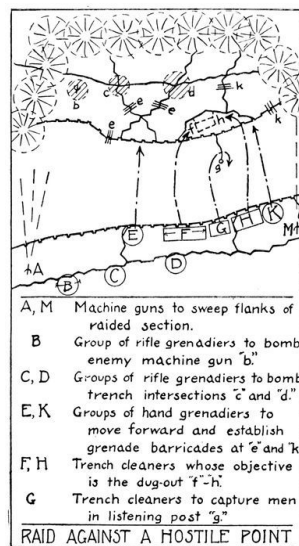
Major 80th Inf.,
Bn. Cmdr.

CHAPTER VIII

SPECIAL OPERATIONS

I. Raids

The object of a raid is to attack a special point in the hostile lines, destroy it, and return with prisoners or information. Such a point may be a salient, blockhouse, observation or listening post, etc. The purpose of the raid is both to prosecute the attrition of the enemy and to gather information. A raid is made with or without a previous artillery preparation. Usually, however, the barbed wire entanglements of the enemy are subjected to a short and intense artillery shelling. Whether or not an artillery bombardment is carried out, the tactics of the raid are the same. Success of a raid depends entirely on its preparation, which must be thorough and complete in every detail so that every man in the raid knows his part.



Preparation. The troops employed for a raid are a specially chosen unit, a platoon or half-company. Better still, a regiment may have a detachment specially trained in this work. When a raid has been decided upon against a particular point in the hostile line, the leader of the raid has the following duties to perform:

1. He goes to the command post of the center of resistance opposite to the point in the enemy's line against which the raid is to be made. Here he gathers all necessary information concerning the enemy by studying maps

and sketches, and comparing them with the ground. He also speaks with occupants of the first line for the purpose of getting information; and calls upon the commander of the center of resistance to send special listening parties, if necessary, to the enemy's line, observing necessary precautions to prevent the enemy from becoming suspicious of a contemplated raid.

From the detailed information, maps and sketches that the leader of the raid has obtained from the headquarters of the center of resistance and from his own reconnaissance, he lays out a model of the hostile point to be attacked. This he does in the rear of his own sector. The model is partially dug in so that the raiding detachment will have an exact duplicate of the enemy's lines to be raided to practice upon.

He now prepares the order for the execution of the raid, which contains the following points:

1. **Purpose of the Raid.** To destroy such and such a dugout, machine gun emplacement, etc., to make prisoners, to gather information, or to prevent a contemplated gas attack on the part of the enemy.

2. **Troops Employed.** Number and designation of the different groups of grenadiers and trench cleaners to be used, equipment of same, and ammunition carried.

3. **Objective of the Raid.** The particular part of a trench, post, blockhouse, etc.

4. **Precise Rôle of Each Group.** Disposition of the groups before the raid in their own lines. This may be in the firing line or in "No Man's Land" in a shell hole reached by crawling at night.

Time and signal of departure.

Means of crossing the enemy's barbed wire entanglements.

Points of the hostile line upon which each group is to make its attack.

Rôle of each group during the raid. Protection of the flank, establishing barricades, proper execution of the raid.

Time or signal to return.

5. Co-operation of the infantry and artillery supporting the raid.

6. Supply of the attacking troops—grenade depots in the firing line.

In practicing the raid in the rear, on the model, the detachment must repeat the execution of the raid at least five times. All the details of the raid

are foreseen and provided for. Each man is made thoroughly proficient in his particular part in the raid, so that, in the confusion and darkness, they will be able to reach their objective and carry out their particular function. Each chief of group must be thoroughly conversant with his duties. The leader of the raid personally conducts the most important of these groups.

Execution of the Raid. The most important element of the execution of a raid is speed, its time of duration being only from five to ten minutes.

Time. If all the points in the enemy's line to be attacked are thoroughly known and located, the raid can be carried out during the night. But if there is some lack of knowledge of the above mentioned points, the raid must be carried out at dusk or dawn, so that the objectives can be at least partly seen.

Disposition of the Groups to Cross the Barbed Wire; Disposition of the Troops before the Raid. The distance of a raid should not be more than fifty yards. For this purpose, the different groups must get into position such a distance from the enemy's lines either by silently crawling across "No Man's Land" or by profiting by a short preliminary bombardment of a few minutes which will prevent the enemy's sentinels and watchers from perceiving them. Their disposition is usually made by hiding or crouching in shell holes.

Destruction of the Barbed Wire. The different ways of destroying barbed wire have been spoken of under Surprise Attacks in the chapter on Defense of a Position.

Signal of Departure. Three methods can be used; previous designation of the hour, use of a whistle, or by means of a rocket. Of these, the last named is probably the best.

There are two different kinds of groups in the raiding party.

The flank groups, armed with hand grenades, protect the groups executing the raid proper by shutting off all avenues of approach of the enemy. Barricades are established by these groups by throwing grenades into the trenches leading to the attacked area.

Between the flank groups, are the groups that carry out the proper mission of the raid. They are armed with incendiary grenades, pistols, and trench knives. Their function is to accomplish any particular destruction of their mission and to capture prisoners.

Infantry Supporting the Raid. The riflemen of the garrison of the center of resistance stand ready at the firing parapets to receive the groups returning from the raid. The machine guns carry out an intense fire on the flanks of the point raided to prevent the approach of hostile reinforcements. The rifle grenadiers of the garrison execute fire upon special points on the flanks or on the rear of the raided area, such as machine gun emplacements, junction of boyaux, etc.

Support of the Artillery. If artillery is used in conjunction with the raid, it may be employed before, during, or after the raid, or in any combination of these periods.

During the raid, two kinds of fire are carried out, that of demolishing the barbed wire entanglements, and special elements of the hostile position that will hinder the raid. This is carried out by the light artillery and trench mortars, principally the latter.

During the raid, the support artillery establishes protective curtains of fire on both flanks and on the rear of the raided portion of the enemy's line. This is to neutralize the enemy's defenses and to cut the area off from communications or reinforcements. In other words, the artillery will isolate by its fire the point of attack of the raid.

At the designated time for the return of the raid or at the signal of a rocket sent up by the raiding party on reaching their own first line, the support artillery establishes its usual defensive barrage in "No Man's Land." It may also fire on the enemy's position with shrapnel, because at this time reinforcements are usually coming up into the raided area. This latter fire may be repeated two or three times during the ensuing hour.

II. Gas Warfare

Gas attacks may be carried out in two ways, by asphyxiating shells or by waves of gas.

The shelling of a position with asphyxiating shells is made to neutralize the action of its defenders and, if possible, to kill and wound some of them. Asphyxiating shells are used in counter-battery fire against the enemy's artillery emplacements. They are also used for shelling a woods in which troops are located and against strong points of the hostile position, either during an offensive period or to carry out attrition of the enemy. Such shelling is also carried out for an offensive, either in the preparation or for the protection of the flanks during the attack. Asphyxiating shells may be thrown on a village or strong point in rear or on the flank of the enemy's position where he may organize a counterattack.

Gas waves are used against the enemy's position with or without following up with the infantry for a raid or an attack. Usually the liquified gas is brought to the first lines in cylinders under heavy pressure. These cylinders must be properly protected from hostile fire until the time that they are to be used. With a favorable wind, the gas is carried across "No Man's Land" and into the enemy's position. Gas waves, of course, are more efficient than asphyxiating shells, but the use of the former method is more difficult as it takes special apparatus and a trained personnel, and can be used only under favorable atmospheric conditions.

Protections against Gas. In general, protective measures against gas waves are the same as against asphyxiating shells. The individual protective apparatus is the gas mask. The essential part of a mask is composed of several layers of porous material containing a chemical that will neutralize the poisonous gas. The gas mask has an efficiency of several hours. Each man carries a gas mask and there are also a certain number kept in reserve in the different depots of the sector. Each gas mask must be properly adjusted and it is the duty of each officer and non-commissioned officer to ascertain that each man in the company has a gas mask in good condition. He must always carry the same.

Each man is trained to put on his mask rapidly. Standing orders compel the man to put on his gas mask immediately the alarm is given or upon his detection of gas by color or odor. Frequent exercises are held to perfect the men in putting on their gas masks rapidly. This drill may be held at the same time as the “stand to” exercises are carried out. The men must have confidence in their gas masks. For this purpose, each man is sent through a “chlorine chamber” several times to give him confidence that, although the gas may be of great density, it has no disagreeable effects.

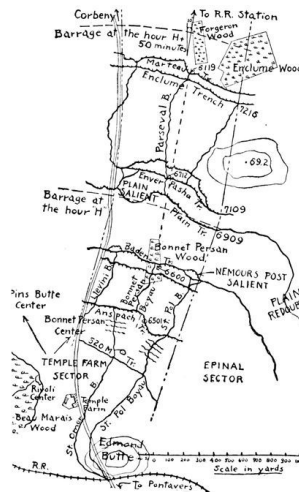
There are also collective protective apparatuses used in dugouts. Each large shelter possesses such apparatus. Besides this, the entrances and openings of each dugout will be closed by a double canvas impregnated with a special chemical solution to neutralize the gas.

Special Precautionary Measures. First, there is established a system of bells, sirens, and klaxons which are used by the watchers to give the signal of alarm. This signal is taken up and repeated in all parts of the position and to the rear. Careful observation on the part of a garrison may detect signs of a coming gas attack. Metallic noises in the hostile line may indicate the transportation of gas cylinders. Very often the enemy will use little balloons to ascertain the velocity and direction of the wind. Foggy weather is a special time of danger.

In each sector or regiment, an officer is in charge of the maintenance of protective gas apparatus and all the measures against gas attack. With the proper precautionary measures and application of the protective measures, the gas attack is not very dangerous.

III. Liquid Fire

Protection against Liquid Fire. The enemy may make an assault with a special detachment of liquid fire operators in the first line. The object of these operators is to throw jets of liquid fire on the occupants of the first line to allow the assaulting columns to penetrate into the position without losses. The only means of combating such an attack is to evacuate the part of the trenches against which the liquid fire is directed, and move by the flanks. Machine guns and automatic rifles should be established to bring flanking fire to bear against the detachment of liquid fire operators. It is impossible to withstand a liquid fire attack if the operators succeed in coming within sixty yards before the garrison can man its parapets.



IV. Mines

We will consider mines only from the point of view of defense by infantry against them. This consideration will therefore have no relation to mine warfare or the construction of countermines by engineers.

Let us consider a concrete example where the enemy is about to explode a mine under your first line of trenches (examine accompanying diagram, Precautions against a Mine). When it has been definitely discovered that the enemy has a mine gallery under your first line and there is no means of combating it by countermining, etc., the infantry in the position must take certain precautions. By listening devices the powder chamber is located. The extent of the crater is also calculated. A new first line is constructed behind, excluding the mine crater and at a distance of about 30 yards from its lip. This re-entrant angle of the first line is made by using boyaux on the flank if they exist. A cover trench is also constructed behind this first line. The infantry moves back to this new line behind the threatened area of explosion. Every other defensive precaution is taken against the explosion and the accompanying assault of the enemy.

Heavy bombing posts are located on the flanks of the crater. The position of these posts is usually at the point of change of direction of the old firing line with the new one of the re-entrant angle. Dugouts may be constructed close to these bombing posts for the protection of the grenadiers during the explosion. Immediately the explosion is over, these grenadiers man their posts and establish a heavy barricade between the posts and the crater by using hand grenades. If the enemy attempts to enter the crater, these grenadiers can make it untenable by throwing hand grenades into it.

Automatic rifle or machine gun emplacements are also located on both flanks of the crater. Their function is to establish flanking fire on hostile troops attempting to approach the crater.

Rifle grenadiers are stationed in the line of trenches to establish a defensive rifle barrage out in front of the crater.

The riflemen will usually occupy the near and flank lips of the crater. They will not occupy the lip of the crater nearest the enemy until it is ascertained that the enemy has no more galleries.

A rocket post is located in the first line on the flanks of the crater. As soon as the crater is exploded, a rocket is sent up from this post calling upon the artillery for a defensive barrage. Usually, special artillery is detailed for this extra fire. It is a reinforced fire, or a combination of preventive fire and protective barrage. It is established on the enemy's first line rather than in "No Man's Land" as a purely defensive barrage.

With all these precautions taken, the infantry await the explosion of the mine.

A Selection from the Catalogue of
G. P. PUTNAM'S SONS



Complete Catalogue sent on application

IT IS THE REAL STUFF

OVER THE TOP

BY AN AMERICAN SOLDIER WHO WENT

ARTHUR GUY EMPEY

MACHINE GUNNER, SERVING IN FRANCE

AUTHOR OF

“FIRST CALL”

For a year and a half, until he fell wounded in No Man's Land, this American soldier saw more actual fighting and real warfare than any war correspondent who has written about the war. His experiences are grim, but

they are thrilling and lightened by a touch of humor as original as the Soldiers Three. And they are true.

*12°, 16 Illustrations and Diagrams, \$1.50 net,
By mail, \$1.60*

TOGETHER WITH TOMMY'S DICTIONARY OF THE TRENCHES

*“Over The Top with the Best of
Luck and Give Them Hell!”*

The British Soldier's War Cry, as he goes over the top of the trench to the charge

FIRST CALL

BY
ARTHUR GUY EMPEY

**GUIDE
POSTS TO
BERLIN**

Author of “OVER THE TOP”

12°. Illustrated. \$1.50 (By mail, \$1.65)

In the amazingly vivid and simple way that has made ***Over the Top*** the most widely read and talked of book in America, and the most successful war book in all history, Empey tells the new soldiers

What they want to know

What they ought to know

What they'll have to know

and what their parents, sweethearts, wives, and all Americans, will want to know, and can do to help.

A practical book by an American who has been through it all.

The chapters headed “Smokes” and “Thank God the Stretcher Bearers” will stand among the war classics.

Here is advice, here are suggestions, overlooked in other books, that will safeguard our boys in France.

The Making of a Modern Army And Its Operations in the Field

A Study Based on the Experience of
Three Years on the French Front

1914–1917

René Radiguet
Général de Division, Army of France

Translated by

Henry P. du Bellet
Formerly American Consul at Rheims

Illustrated

The younger Americans who are now in training for active service in the field, and particularly those who have secured commissions as officers or who are preparing to compete for such commissions, will have a very direct interest in the instructions and suggestions presented by General Radiguet in regard to the organization of an army and the method of its operations in the field. General Radiguet's treatise is based upon a varied experience in the campaigns of the present war.

The old text-books must be put to one side. The methods of organization and the methods of fighting have alike changed. It is only those who have had responsibilities as leaders in the present war whose instructions can be accepted as authoritative.

G. P. Putnam's Sons

New York

London



TRANSCRIBER'S NOTE

1. Silently corrected typographical errors and variations in spelling.
 2. Retained anachronistic, non-standard, and uncertain spellings as printed.
-

***** END OF THE PROJECT GUTENBERG EBOOK TACTICS AND
DUTIES FOR TRENCH FIGHTING *****

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the

collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with

this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official

version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.

- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES

EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a)

distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact

links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility: www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.